

Distribution-independent SQ learning does not imply low dimension complexity

A negative answer to Open Question 2 of Feldman, Kamath and Srebro

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Abstract

Distribution-independent statistical-query learning does not imply low dimension complexity, even for nondegenerate probabilistic variants. If every product distribution has a monochromatic rectangle of mass at least $1/\rho$, a common learner uses $O(\rho(\log |H| + \log(1/\epsilon)))$ queries at tolerance $\Omega(\epsilon/(\rho \log(1/\epsilon)))$. For random incidence flips on $2M$ points, this yields $O_\epsilon(\log M)$ queries despite dimension $\Omega(M^{1/3}/\log M)$, exact probabilistic dimension $\Omega(M^{1/3}/\log^2 M)$, and averaged probabilistic dimension $\Omega_\eta(M^{1/3}/\log M)$ for fixed $\eta < 1/2$. Proper deterministic implementations run in time polynomial in the supplied table. Under subexponential pseudorandom-function security, a key-known polynomial-time learner uses $O_\epsilon(n^2)$ queries while almost every key gives dimension at least $n^c/(30 \log_2 n)$, for any prescribed fixed c and $n = \log_2 |X|$. These classes also admit engineered polynomial-size differentiable-learning simulations. Bounded product-average communication is sufficient for SQ learning, but cube halfspaces show it is unnecessary. The prescribed Gaussian-initialized, bias-free online-SGD question remains open.

1 Introduction

Feldman et al. [17, Open Question 2] ask:

Open Question 2 (SQ Learning vs. Dimension Complexity) Is there a constant C such that for every class $\mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$, over any domain $\mathcal{X}$, and any $\epsilon < 1/4$, if there exists an (m, τ) -SQ algorithm s.t. for every input distribution $\mathcal{D}$ and every $h^* \in \mathcal{H}$, the algorithm returns a predictor $\hat{h}$ with $\mathbb{E}\mathcal{L}_{\mathcal{D}, h^*}(\hat{h}) \leq \epsilon$ (expectation over the randomness of the algorithm), then $\text{dc}(\mathcal{H}) \leq C \cdot m/\tau^2$.

The answer is negative. One learner can serve every input distribution with few statistical queries while every common linear representation requires exponentially many dimensions in the domain's encoding length.

Here is the quantitative question in the notation used below. Let X be finite, $n = \log_2 |X|$, and $H \subseteq \{-1, +1\}^X$. Let $\text{dc}(H)$ be the least dimension of a strict linear representation of every $h \in H$ by one feature map. Suppose a randomized adaptive (m, τ) statistical-query learner has expected classification error at most a fixed $0 < \epsilon < 1/4$ for every marginal D on X and target $h \in H$. The proposed conclusions, in decreasing strength, are

$$\text{dc}(H) \leq Cm/\tau^2, \tag{L} \tag{1.1}$$

$$\text{dc}(H) \leq p(m, 1/\tau), \tag{P} \tag{1.2}$$

$$\text{dc}(H) \leq p(m, 1/\tau, n), \tag{Pn} \tag{1.3}$$

where C and the polynomial p are universal. We also consider $\text{dc}_{1/2}$, which permits a random embedding realizing each target exactly with probability at least $1/2$, and dc^η , which permits expected best-linear-predictor error at most η . Their precise quantifiers and tie conventions appear in Section 2. Each proposed conclusion fails for these notions as well.

Main Theorem. *For every finite nonempty class H , write*

$$\text{rect}(H) = \inf_{\mu \in \Delta(X), \nu \in \Delta(H)} \max_{S \times T \text{ monochromatic}} \mu(S)\nu(T).$$

If $\text{rect}(H) \geq 1/\rho$ with $\rho \geq 1$, then for every $0 < \epsilon < 1/4$ one distribution-independent SQ learner achieves expected error at most ϵ against every valid adaptive oracle, using

$$m = O(\rho(\log |H| + \log(1/\epsilon))), \quad \tau = \Omega\left(\frac{\epsilon}{\rho \log(1/\epsilon)}\right).$$

For every prescribed fixed $0 \leq \eta < 1/2$ and infinitely many M , there are classes on $2M$ points with $\text{rect}(H) \geq 2^{-15}$ and

$$\text{dc}(H) = \Omega\left(\frac{M^{1/3}}{\log M}\right), \quad \text{dc}_{1/2}(H) = \Omega\left(\frac{M^{1/3}}{\log^2 M}\right), \quad \text{dc}^\eta(H) = \Omega_\eta\left(\frac{M^{1/3}}{\log M}\right)$$

with constants allowed to depend on η . They have proper deterministic learners with $m = O_\epsilon(\log M)$ and $\tau = \Omega_\epsilon(1)$, computable in polynomial time from the supplied table.

The parameter constants are given in Lemma 3.2; the separation follows from Theorems 4.1, 4.2, 5.1 and 5.2 and Lemma 4.3. The deterministic implementation has a pointwise worst-case error guarantee, so the counterexample does not depend on averaging the learner's error over its random seed.

Mechanism. A fixed feature map must realize every example-hypothesis sign simultaneously. A learner can first ask where the unknown marginal places mass, then label an agreed region or remove inconsistent hypotheses. A finite minimax argument turns the product-distribution rectangle property into a distribution over rectangles that covers every point without knowing the marginal. RECTIFY charges either action to the product of the rectangle's two masses. One potential controls hypothesis eliminations; another controls peeled mass. Their combination yields the finite query budget. The incidence-flip construction preserves this rectangle structure while counting forces high sign-rank.

Contributions.

- A rectangle-based sufficient condition for distribution-independent SQ learning, with explicit tolerance and a worst-case query bound (Theorem 3.1 and Lemma 3.2).
- A negative answer to (L), (P) and (Pn), including exact and averaged probabilistic dimension under nondegenerate loss conventions (Corollary 4.4).
- Table-polynomial cap samplers, a deterministic implementation, and proper learning for the separating family (Theorems 5.1 and 5.2).
- A succinct order-median learner with $O(n^2)$ queries, and conditional key-known polynomial-time separations for every prescribed polynomial degree (Theorems 5.4 and 6.2 and Corollary 6.5).
- Communication sufficient conditions, spectral obstructions, and examples distinguishing rectangle size from SQ complexity (Theorems 7.1, 7.3 and 7.9).
- Quantified marginal adaptation and resource barriers, together with a precise engineered-gradient corollary and the remaining online-SGD question (Theorems 7.10 and 10.1, Corollary 8.1, and Propositions 10.2 and 10.3).

Imported ingredient	Role in the argument
Incidence flips and rectangle geometry [24]	Supply a class with large rectangles and many freely chosen signs.
Warren’s theorem [5, 33]	Counts strict low-rank sign patterns; the application-specific constants and probabilistic reductions are derived here.
Homogeneous rectangles [4, 24]	Sharpen the constant to 2^{-15} ; Proposition L.1 independently gives 2^{-18} .
Finite minimax [24]	Supplies the mixture device; the learning progress and oracle analysis use that device.
SQ and differentiable-learning characterizations [2, 15]	Used only for the separate comparisons in Section 8 and Appendix I.

Table 1: Imported results and their roles. The adaptive learner, its proper and succinct implementations, and the stated reductions are proved in the sections indicated.

Roadmap. Section 3 proves the learner; Section 4 gives the separating family. Sections 5 and 6 treat implementation and succinct descriptions. Section 7 compares sufficient conditions and certificates. Section 8 addresses differentiable learning, and Section 10 states the remaining questions. Deferred proofs and reproducibility material follow the references.

Scope. SQ query complexity charges neither the class description nor preprocessing. Table-polynomial and seed-polynomial running times are distinguished throughout. A high-probability statement about keys or advice does not select an efficient deterministic sequence of good strings. The conditional statements assume the specified PRF security. Probabilistic dimensions use Boolean predictions with an explicit tie rule; a raw-score loss that declares every zero score correct is excluded. The original class-dependent Gaussian-initialized online-SGD implication remains open.

2 Preliminaries and notation

All domains and hypothesis classes are finite and nonempty. The logarithm $\log_2$ is base two and $\ln$ is natural. Unsubscripted logarithms in asymptotic bounds may use either base. Constants in O_ϵ or Ω_ϵ may depend on a fixed accuracy ϵ , but not on the class or its size. Repeated rows are permitted when a geometric construction indexes identical hypotheses separately.

For a marginal $D \in \Delta(X)$ and Boolean functions h, g , define

$$L_{D,h}(g) = \sum_x D(x) \mathbf{1}_{g(x) \neq h(x)}, \quad \mathbb{E}_D[hg] = 1 - 2L_{D,h}(g).$$

A statistical query is a function $q : X \times \{-1, +1\} \rightarrow [-1, 1]$. At tolerance $\tau > 0$, its response may be any v satisfying $|v - \mathbb{E}_D q(x, h(x))| \leq \tau$ [28]. An oracle policy may choose its answers using the entire public query-answer history, including earlier queries that reveal previous random choices and the current query. This is the strongest nonanticipating adaptive-policy reading of the source’s arbitrary valid responses; the policy is not given unrevealed future private coins. A randomized learner must satisfy

$$\forall D \forall h \in H \forall \text{ valid adaptive oracle policies } O, \quad \mathbb{E}_\omega L_{D,h}(A_\omega^O) \leq \epsilon. \quad (2.1)$$

The expectation is over the learner’s private randomness only. A deterministic learner is a real-response decision tree of depth at most m ; a randomized learner is a distribution over such trees. Early stopping can be padded by zero queries. For our finite-state algorithms, all output losses and events are measurable. Randomized oracle policies can be handled by conditioning on their randomness. For computational statements, the oracle returns a finite binary encoding at the stated precision; reserving half the tolerance for rounding gives this interface from a finer-precision answer. Query-complexity statements permit arbitrary real responses and charge no answer-reading cost.

Symbol	Meaning
X, H, K, n	Domain, indexed class, $K = H $, $n = \log_2 X $.
D, μ, ν	Unknown example marginal; auxiliary point and hypothesis priors.
m, τ, ϵ	Query budget, additive tolerance, target classification error.
$U, V, S \times T$	Unpeeled points, surviving hypotheses, sampled monochromatic rectangle.
$r = 1/\rho$	A known lower bound on $\text{rect}(H)$.
N, M, I	Grid scale, $M = N^3$, and number of independently flippable incidences.
$\text{dc}, \text{dc}_{1/2}, \text{dc}^\eta$	Strict, exact probabilistic, and averaged probabilistic dimensions.
Γ, Λ, R	Certificate Gram matrix, its operator norm, and the ratio K/Λ .

Table 2: Notation. The round-budget symbol R in Lemma 3.2 is local to RECTIFY and is separate from the spectral ratio.

Dimension complexity measures simultaneous linear representability [7]. We use its strict convention: $\text{dc}(H)$ is the least integer $d \geq 0$ for which one map $\phi : X \rightarrow \mathbb{R}^d$ satisfies

$$\forall h \in H \exists w_h \in \mathbb{R}^d \forall x \in X, \quad h(x) \langle w_h, \phi(x) \rangle > 0. \quad (2.2)$$

A rank factorization of the truth table is exactly such a representation, so $\text{dc}(H)$ equals its sign-rank. The minimum exists because the coordinate embedding $x \mapsto e_x$ works. Allowing a fixed tie rule changes the ordinary dimension by at most one, by Lemma 2.3, so the asymptotic separations cover that convention as well.

For the probabilistic notions, set $\text{sign}(0) = +1$ and classify the score before evaluating loss. For $0 < \delta < 1$, the exact probabilistic dimension $\text{dc}_\delta(H)$ is the least d admitting one law on embeddings such that

$$\forall D, h, \quad \Pr_\phi[\exists w : L_{D,h}(\text{sign}\langle w, \phi \rangle) = 0] \geq 1 - \delta. \quad (2.3)$$

The case $\delta = 1/2$ is used below. This is the probabilistic representation notion associated with Chornomaz et al. [10]. The averaged dimension of Kamath et al. [26], denoted here by $\text{dc}^\eta(H)$, is the least d admitting one embedding law with

$$\forall D, h, \quad \mathbb{E}_\phi \inf_w L_{D,h}(\text{sign}\langle w, \phi \rangle) \leq \eta. \quad (2.4)$$

The embedding law precedes both D and h ; the selected vector w need not. For each fixed embedding and marginal, the infimum is a minimum because there are only finitely many Boolean prediction patterns on X .

Zero scores. The raw-score formulas in Feldman et al. [17, Section 4], read together with their strictly negative-product loss, assign zero loss to $w = 0$ for every class. In particular, $\Pr[h(x) \langle w, \phi(x) \rangle < 0] = 0$ when $w = 0$. That formula therefore gives a vacuous probabilistic dimension unless the prediction or tie rule is specified. Equations (2.3)–(2.4) use Boolean predictions. Lemma 2.3 establishes the same asymptotic lower bounds for the opposite tie rule and for a convention counting zero scores as errors. These are the nondegenerate probabilistic formulations refuted here.

A monochromatic rectangle is a product $S \times T \subseteq X \times H$ on which $h(x)$ is a fixed color $c \in \{-1, +1\}$. Put

$$\text{rect}(H) = \inf_{\mu \in \Delta(X), \nu \in \Delta(H)} \max_{S \times T \text{ monochromatic}} \mu(S) \nu(T). \quad (2.5)$$

Restricting to a nonempty submatrix cannot decrease this quantity: extend its two distributions by zero. Hence every lower bound $r \leq \text{rect}(H)$ remains valid after peeling points or eliminating rows.

Lemma 2.1 (Finite minimax and pointwise coverage). *Let $\mathcal{A}$ be a nonempty finite action set and Z a nonempty finite set of instance types. Let $g_a \in \mathbb{R}^Z$ and $r \in \mathbb{R}$. The following statements are equivalent:*

$$\forall \mu \in \Delta(Z), \max_{a \in \mathcal{A}} \mathbb{E}_\mu g_a \geq r; \quad \exists \lambda \in \Delta(\mathcal{A}), \forall z \in Z, \mathbb{E}_\lambda g_a(z) \geq r.$$

The result also applies when an arbitrary instance set induces only finitely many payoff vectors $(g_a(z))_{a \in \mathcal{A}}$.

Proof. The reverse implication follows by averaging. For the forward direction, let $C = \text{conv}\{g_a : a \in \mathcal{A}\}$ and $Q = r\mathbf{1} + \mathbb{R}_{\geq 0}^Z$. If disjoint, their minimum distance is positive and attained: C is compact, and a bounded-distance minimizing sequence has bounded Q -components. Let (c, z) attain it and put $v = z - c \neq 0$. Differentiation along feasible segments gives

$$\langle v, c' - c \rangle \leq 0 \quad (c' \in C), \quad \langle v, z' - z \rangle \geq 0 \quad (z' \in Q).$$

Taking $z' = z + te_x$ yields $v_x \geq 0$. Moreover,

$$\max_{c' \in C} \langle v, c' \rangle \leq \langle v, c \rangle < \langle v, z \rangle = \min_{z' \in Q} \langle v, z' \rangle = r \sum_x v_x.$$

Normalize v to a probability distribution, contradicting the first condition. Hence $C \cap Q \neq \emptyset$, and its convex coefficients give λ . For finitely many payoff types, retain one representative of each type and apply the finite result. $\square$

Lemma 2.2 (From uniform rectangles to product weights). *Suppose a property of finite matrices is preserved by repeating rows and columns. Suppose every matrix with that property has a monochromatic rectangle with row fraction at least a and column fraction at least b . Then every such matrix has rectangle ratio at least ab under arbitrary product distributions.*

Proof. For strictly positive rational weights, repeat rows and columns according to common-denominator multiplicities. Projecting a monochromatic rectangle back to the original indices preserves its signs and can only increase its weighted side masses. For arbitrary weights, approximate by strictly positive rational distributions. The largest rectangle mass is the maximum of finitely many continuous products of linear functions of the weights, hence is continuous. Passing to the limit gives the assertion. $\square$

Lemma 2.3 (Tie conventions and strict realization). *For any finite family of predictions obtained from a d -dimensional embedding with the convention $\text{sign}(0) = +1$ or -1 , the same Boolean prediction matrix has a strict realization in dimension at most $d + 1$. If all zero scores are counted as errors, replacing their predictions by $+1$ cannot increase loss, for any marginal and target.*

The proof, including the attained-infimum and zero-incidence cases, appears in Appendix A.

3 The rectangle learner

Theorem 3.1 (Distribution-independent rectangle learner). *Let X, H be finite and nonempty, $K = |H|$, and suppose $\rho \geq 1$ and $\text{rect}(H) \geq 1/\rho$. For every $0 < \epsilon < 1/4$, one randomized algorithm satisfies (2.1) with*

$$m = O(\rho(\log K + \log(1/\epsilon))), \quad \tau = \Omega\left(\frac{\epsilon}{\rho \log(1/\epsilon)}\right).$$

Query count is a worst-case bound. The learner is explicit from the truth table and a valid lower bound on $\text{rect}(H)$; the general implementation may solve exponential-size linear programs.

Algorithm 1: RECTIFY: rectangle peeling and elimination

Input: A table H , a bound $r \leq \text{rect}(H)$, and $0 < \epsilon < 1/4$.

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1 Set  $U \leftarrow X$ ,  $V \leftarrow H$ , and use  $P, \tau, R$  from Lemma 3.2;
2 for  $t = 1, \dots, R$  do
3   if  $U = \emptyset$  then
4     return the partial predictor;
5   Query  $\hat{u} \approx D(U)$ ;
6   if  $\hat{u} \leq \epsilon/4$  then
7     return the partial predictor, extended by +1 on  $U$ ;
8   With  $\nu$  uniform on  $V$ , draw  $(S, T, c)$  from Lemma 3.3;
9   Query  $\hat{s} \approx D(S)$ ;
10  if  $\hat{s} < (r/2)\hat{u}$  then
11    Reject this proposal and continue;
12  Query  $\hat{w} \approx D(S \cap \{h^* \neq c\})$ ;
13  if  $\hat{w} \leq 2\tau$  then
14    Set the predictor to  $c$  on  $S$ ;  $U \leftarrow U \setminus S$ ;
15  else
16     $V \leftarrow V \setminus T$ ;
17 return the partial predictor, extended by +1 on  $U$ ;
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Lemma 3.2 (Exact parameters). *For a finite nonempty H , $\rho \geq 1$, $\text{rect}(H) \geq 1/\rho$, and $0 < \epsilon < 1/4$, Theorem 3.1 holds with*

$$r = 1/\rho, \quad P = 1 + \lceil 4\rho \ln(8/\epsilon) \rceil, \quad \tau = \epsilon/(24P), \quad (3.1)$$

$$B = \ln K + \ln(8/\epsilon) + 1, \quad R = \lceil 8\rho(B + \ln(2/\epsilon)) \rceil, \quad m = 3R + 1. \quad (3.2)$$

Lemma 3.3 (A marginal-free rectangle mixture). *Assume H is finite, $0 < r \leq \text{rect}(H)$, and fix nonempty $U \subseteq X$, nonempty $V \subseteq H$, and a distribution ν on V . There is a distribution λ over monochromatic rectangles $S_R \times T_R \subseteq U \times V$ such that*

$$\forall x \in U, \quad \mathbb{E}_{R \sim \lambda}[\mathbf{1}_{S_R}(x)\nu(T_R)] \geq r. \quad (3.3)$$

This distribution can be computed from (H, U, V, ν) without a query about D .

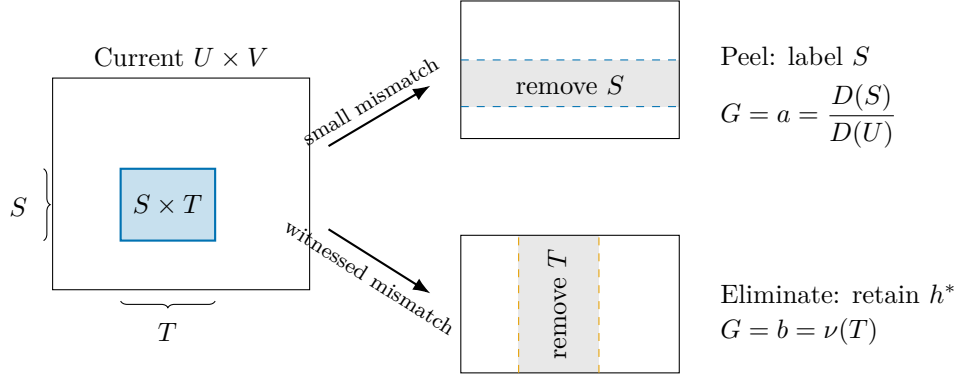
Proof. For each rectangle Q , set $z_Q(x) = \mathbf{1}_{S_Q}(x)\nu(T_Q)$. The rectangle hypothesis on the restricted matrix gives $\max_Q \langle \mu, z_Q \rangle \geq r$ for every $\mu \in \Delta(U)$. Apply Lemma 2.1. The resulting feasibility problem has rational coefficients when ν, r are rational, so it admits a rational solution. No information about D enters this computation. $\square$

Proof idea. A sampled rectangle either labels a region or removes hypotheses. Both outcomes make progress at least the product ab of the two normalized masses. Pointwise coverage gives a lower bound on its conditional expectation. The target's weight and the unpeeled mass provide a deterministic upper bound on total progress, yielding a finite query budget. Figure 1 depicts the two outcomes.

Peeled sets are disjoint. Eliminations leave U unchanged. The cap construction in Section 5.1 supplies an alternative implementation of the mixture.

Proof of Theorem 3.1 and Lemma 3.2. Fix D, h^* and any valid oracle policy. First,

$$\tau \leq \epsilon/8, \quad \tau \leq r\epsilon/64. \quad (3.4)$$



Either accepted action has $G \geq ab$; pointwise coverage forces $\mathbb{E}[G] \geq r/4$.

Figure 1: One RECTIFY proposal. The horizontal axis indexes surviving hypotheses and the vertical axis unpeeled points. A nearly monochromatic region is labeled and removed; a disagreement removes the rectangle's hypotheses. The two actions are charged to different potentials, but both dominate the product ab (Theorem 3.1).

Indeed $P \geq 4\rho \ln(8/\epsilon) > 4\rho \ln 32$, and $24 \cdot 4 \ln 32 > 64$. At a nonstopping round, write

$$u = D(U) > \epsilon/4 - \tau \geq \epsilon/8, \quad a = D(S)/u, \quad b = \nu(T), \quad w = D(S \cap \{h^* \neq c\}).$$

Thus $\tau/u \leq r/8$. If the proposal is accepted, validity implies

$$D(S) \geq (r/2)u - (1 + r/2)\tau \geq (5r/16)u \geq (r/4)u.$$

Conversely, every rectangle with $a \geq 3r/4$ is accepted, since

$$\hat{s} - (r/2)\hat{u} \geq (r/4)u - (1 + r/2)\tau \geq (r/16)u > 0.$$

These inequalities hold at the extreme valid oracle answers, not only for exact answers.

If $h^* \in T$, monochromaticity gives $w = 0$, hence $\hat{w} \leq \tau < 2\tau$. Therefore the algorithm never eliminates the target. In particular, V remains nonempty. On an elimination, its normalized target weight increases by a factor $1/(1-b)$. Because $-\ln(1-b) \geq b$ for $0 \leq b < 1$, summing gives

$$\sum_{\text{eliminations}} b \leq \ln K. \quad (3.5)$$

For completeness, $\ln t \leq t - 1$ for $t > 0$ follows by minimizing $t - 1 - \ln t$, whose derivative is $1 - 1/t$. Taking $t = 1 - b$ proves the displayed inequality. The target weight starts at $1/K$ and never exceeds one.

Each peel multiplies u by at most $1 - r/4$. Immediately before every peel, $u \geq \epsilon/8$. Before the j th peel,

$$\epsilon/8 \leq (1 - r/4)^{j-1} \leq e^{-(j-1)r/4}.$$

Hence there are at most P peels. Their combined wrong-color mass is at most $3P\tau = \epsilon/8$, since a peeling answer implies $w \leq 3\tau$. At a normal stop the residual mass is at most $\epsilon/4 + \tau$, or zero. The error at such a stop is at most $\epsilon/2$.

It remains to bound the probability of reaching the round limit. Define a round's progress by $G = a$ for a peel, $G = b$ for an elimination, and $G = 0$ for a rejected proposal. Condition on all public history through the current stopping answer $\hat{u}$, but before the fresh rectangle draw. This includes every earlier query and response; conditional on that history the next rectangle has the

prescribed mixture law. Always $0 \leq G \leq 1$. For every accepted rectangle, either action has $G \geq ab$. All rectangles with $a \geq 3r/4$ are accepted. Averaging (3.3) under D restricted to U gives $\mathbb{E}[ab] \geq r$. Consequently,

$$\mathbb{E}[G \mid \text{history}] \geq \mathbb{E}[ab] - \mathbb{E}[ab \mathbf{1}_{a < 3r/4}] \geq r - (3r/4)\mathbb{E}b \geq r/4. \quad (3.6)$$

This holds for either oracle-induced action on every accepted rectangle. The oracle policy was fixed at the start of the proof and may react to each query; no supremum over policies is moved inside an expectation.

On any trajectory of nonstopping rounds, the sum of all peeling a values except the last is at most the logarithm of the reciprocal residual mass immediately before the last peel. Indeed $a \leq -\ln(1-a)$ and the residual ratios telescope. This sum is at most $\ln(8/\epsilon)$; the last a is at most one. If there are no peels the sum is zero. Together with (3.5), this proves

$$\sum G \leq B. \quad (3.7)$$

There is no logarithm of zero: a peel with $a = 1$ must be the last peel.

After any normal termination, extend the sequence by setting $G = 1$. Let $p = r/4$. For $0 \leq g \leq 1$, convexity places e^{-g} below the chord between its endpoint values:

$$e^{-g} \leq 1 - (1 - e^{-1})g.$$

Since $1 - e^{-1} > 1/2$, conditional expectation and (3.6) imply

$$\mathbb{E}[e^{-G} \mid \text{history}] \leq e^{-p/2}.$$

Induction gives $\mathbb{E}e^{-\sum_{t=1}^R G_t} \leq e^{-pR/2}$. On the event of no normal termination by the budget, (3.7) holds. Markov's inequality therefore gives

$$\Pr[\text{budget termination}] \leq e^{B-pR/2} \leq \epsilon/2.$$

A point that becomes empty after the final peel already has the normal-stop error bound; counting it as budget termination only weakens this estimate. Finally, error is always at most one, so expected error is at most $\epsilon/2 + \epsilon/2 = \epsilon$. There are at most three queries per round and at most one extra stopping check. The big- O and Ω bounds follow directly from the ceilings and $\rho \geq 1$, $\epsilon < 1/4$. $\square$

Domain-size consequence of Theorem 3.1. The dependence on $\log K$ can always be bounded in terms of the domain and ρ . If H shatters $v \geq 1$ points, put the uniform marginal on those points and the uniform prior on one representative of each of their 2^v label patterns. A nonempty monochromatic rectangle using s of the points has mass at most $s/(v2^s) \leq 1/(2v)$, because $s \leq 2^{s-1}$ for every positive integer s (by induction). Therefore

$$\text{VC}(H) \leq \rho/2.$$

After removing duplicate rows, the finite counting inequality $K \leq \sum_{i=0}^v \binom{|X|}{i}$ follows by induction on $|X|$: split restrictions on the last point into their union A and intersection B . Their sizes sum to K , the VC dimension of A is at most v , and that of B is at most $v-1$, since shattering a set in B also permits either label on the last point. Pascal's identity completes the induction, with the empty domain as base case. For $v = 0$, all rows are identical and $K = 1$. For $1 \leq v \leq Q = |X|$, setting $t = v/Q \leq 1$ gives

$$t^v \sum_{i=0}^v \binom{Q}{i} \leq (1+t)^Q \leq e^v, \quad \ln K \leq v \ln(eQ/v) \leq (\rho/2) \ln(eQ).$$

Consequently the same learner, with no statistical-dimension conversion, has

$$m = O(\rho^2 \ln(e|X|) + \rho \ln(1/\epsilon)), \quad \tau = \Omega(\epsilon/(\rho \ln(1/\epsilon))).$$

For bounded ρ , this is logarithmic in the domain size at fixed accuracy. The argument is a proved consequence, not a conjectured converse characterizing SQ complexity.

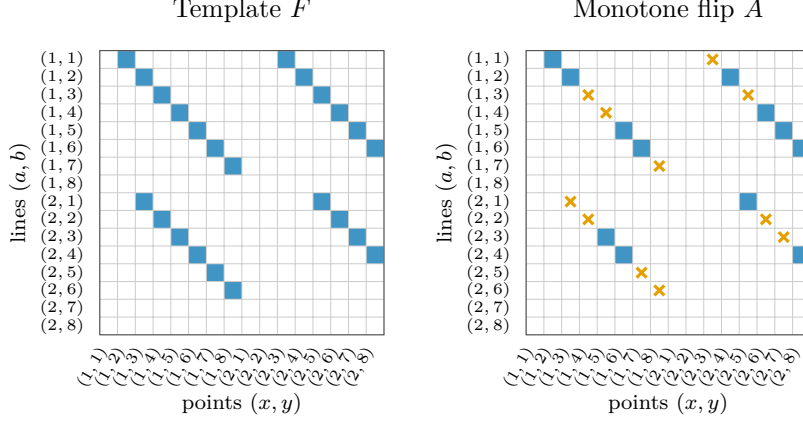


Figure 2: The construction at $N = 2$. Blue cells are negative incidences $y = ax + b$; white cells are positive. The right panel changes a fixed subset of the negative cells to positive, marked by orange crosses. Rows and columns retain their geometric indices. Larger scales supply the counting separation in Theorem 4.1.

4 The separating family

Let $N \geq 2$, $M = N^3$, and $K = 2M$. Points are $p = (x, y) \in [N] \times [2N^2]$. Rows are affine lines $\ell = (a, b) \in [N] \times [2N^2]$. Put $F_{\ell p} = -1$ exactly when $y = ax + b$. A monotone flip matrix A changes any subset of these -1 entries to $+1$ and leaves all other entries unchanged. In the random family the choices are independent fair bits. Write H_A for its row class; figure 2 shows $N = 2$.

Theorem 4.1 (Construction and ordinary dimension). *Every monotone flip A satisfies $\text{VC}(A) \leq 2$ and $\text{rect}(A) \geq 2^{-15}$. The template has $\text{signrank}(F) \leq 6$ and exactly*

$$I = 2N^4 - \left(\frac{N(N+1)}{2} \right)^2 \geq N^4 \quad (4.1)$$

independent flip positions. Set $B_0 = \log_2(4eK)$ and $s_0 = \lfloor I/(4KB_0) \rfloor$. A random A satisfies

$$\Pr[\text{dc}(H_A) \leq s_0] \leq 2^{-I/2}.$$

Every member also satisfies $\text{dc}(H_A) \leq 2N + 1$. In particular most matrices have $\text{dc}(H_A) = \Omega(N/\log M)$. For every $0 < \epsilon < 1/4$, all members have a learner with $m = O_\epsilon(\log M)$ and $\tau = \Omega_\epsilon(1)$, by Theorem 3.1 with $\rho = 2^{15}$. Independently of the sharp rectangle input, the self-contained cap proof in Appendix L supplies $\rho = 2^{18}$ and the same asymptotic learning bound.

The proof uses the strict factorization

$$\langle (x^2, xy, y^2, x, y, 1), (a^2, -2a, 1, 2ab, -2b, b^2 - 1/2) \rangle = (ax + b - y)^2 - 1/2.$$

Positive template rectangles survive every flip. A negative template rectangle has a singleton side, so partitioning the other side by its new labels loses at most half its mass. Independent flips then supply I free signs, to which Warren's counting theorem applies. Appendix B gives the complete proof. Exactly $N(N+1)/2$ template rows have no incidences; all remain positive.

The restriction in (B.2) cannot be omitted. Every sign matrix is a monotone flip of the all-negative, rank-one matrix. More quantitatively, let W be a $k \times k$ Walsh matrix, k a power of two. Its orthogonality gives $\|W\|_{\text{op}} = \sqrt{k}$. A monochromatic rectangle of side sizes a, b satisfies

$$ab = |\mathbf{1}_S^T W \mathbf{1}_T| \leq \sqrt{k} \sqrt{ab},$$

so its uniform product mass is at most $1/k$. For $k > 32$, this contradicts the asserted unrestricted bound $2^{-(2 \cdot 1 + 3)}$.

4.1 Probabilistic dimension

Theorem 4.2 (Approximate counting and exact probabilistic dimension). *Fix $0 \leq \eta < 1/2$ and write*

$$h_2(\eta) = -\eta \log_2 \eta - (1 - \eta) \log_2 (1 - \eta), \quad c_\eta = 1 - h_2(\eta) > 0, \quad s_\eta = \left\lfloor \frac{c_\eta I}{4KB_0} \right\rfloor,$$

with $0 \log 0 = 0$. For the random incidence-flip matrix,

$$\Pr[\text{dc}^\eta(H_A) < s_\eta] \leq 2^{-c_\eta I/2}. \quad (4.2)$$

Consequently most A have $\text{dc}_{1/2}(H_A) = \Omega(N/\log^2 M)$ and $\text{dc}^\eta(H_A) = \Omega_\eta(N/\log M)$.

Proof. Let J_ℓ be the set of template incidences in row ℓ , with $k_\ell = |J_\ell|$. For each nonempty J_ℓ , take D_ℓ uniform on it. If a law on d -dimensional embeddings witnesses $\text{dc}^\eta(H_A) \leq d$, then

$$\mathbb{E}_\phi \sum_{\ell: k_\ell > 0} \frac{k_\ell}{I} \min_w L_{D_\ell, h_\ell}(\text{sign}\langle w, \phi \rangle) \leq \eta.$$

For a fixed embedding, only finitely many Boolean predictions are possible, so each infimum is attained as a minimum of finitely many loss values. There exists one embedding with the displayed weighted sum at most η . Otherwise its positive excess over η would have strictly positive expectation. Choose one minimizing weight vector for each row. Their prediction matrix makes at most ηI mistakes on the incidence positions.

With $\text{sign}(0) = +1$, this prediction matrix has strict sign-rank at most $d + 1$, not necessarily at most d . Add the constant feature one. In each row add a strictly positive bias smaller than the absolute value of every negative score in that row, if there are negative scores. This retains all signs and turns every zero into a positive score. Finiteness makes the bias choice possible. Rows with no incidence may use any weight vector.

For any fixed sign matrix B , the number of possible incidence flip patterns within ηI mistakes is at most

$$\sum_{j=0}^{\lfloor \eta I \rfloor} \binom{I}{j} \leq 2^{h_2(\eta)I}.$$

For $0 < \eta < 1/2$, this follows from

$$1 \geq \sum_{j \leq \eta I} \binom{I}{j} \eta^j (1 - \eta)^{I-j} \geq 2^{-h_2(\eta)I} \sum_{j \leq \eta I} \binom{I}{j};$$

the second inequality uses $j \leq \eta I$ and $\eta/(1 - \eta) < 1$. For $\eta = 0$ the sum is one. Combining with (B.3), the probability that any rank-at-most- s_η sign matrix makes at most ηI incidence errors is at most

$$2^{2Ks_\eta B_0 + h_2(\eta)I - I} \leq 2^{-c_\eta I/2}.$$

An embedding of dimension below s_η would produce just such a strict sign matrix after adding the constant coordinate. If $s_\eta = 0$, the event in question is empty. This proves (4.2). $\square$

Lemma 4.3 (From exact probabilistic to strict dimension). *For every finite nonempty class H with $K = |H|$,*

$$\text{dc}(H) \leq (\lceil \log_2 K \rceil + 1) \text{dc}_{1/2}(H) + 1. \quad (4.3)$$

Resource	Incidence flips on $2M$ points, $M = N^3$
VC dimension; rectangle ratio	$VC \leq 2$; $\text{rect} \geq 2^{-15}$.
Strict dimension, most flips	$\Omega(M^{1/3}/\log M) \leq \text{dc} \leq 2M^{1/3} + 1$.
Exact probabilistic dimension	$\text{dc}_{1/2} = \Omega(M^{1/3}/\log^2 M)$.
Averaged dimension	$\text{dc}^\eta = \Omega_\eta(M^{1/3}/\log M)$ for fixed $\eta < 1/2$.
Table-known learners	$m = O_\epsilon(\log M)$, $\tau = \Omega_\epsilon(1)$; proper and deterministic; table-polynomial time.
Key-known median learner	$m = O_\epsilon(n^2)$, $\tau = \Omega_\epsilon(1)$; deterministic improper for every key; proper for regular stars.
Conditional high dimension	$\text{dc} \geq n^c/(30 \log_2 n)$ for almost every PRF key, each fixed c .
Refuted conclusions	(L), (P), (Pn) for the three nondegenerate dimensions; every prescribed degree for the corresponding succinct families.

Table 3: The separating family’s parameters. The probabilistic lower bounds concern each prescribed fixed $\eta < 1/2$ and hold with exponentially small failure probability. A polynomial-time evaluator with a supplied key is distinguished from deterministic selection of a good key.

Proof. Choose a full-support distribution on X . Zero loss under this distribution means correct labels at every point. Draw $t = \lceil \log_2 K \rceil + 1$ independent embeddings from a law witnessing $\text{dc}_{1/2}(H) = d$. Each fixed target fails in all t embeddings with probability at most 2^{-t} . A union bound gives probability at most $K2^{-t} < 1$ that some target fails everywhere. Hence one collection works for all targets. Concatenate its t embeddings; each row selects a successful block. Adding one constant coordinate as above makes the representation strict. This proves (4.3). $\square$

Corollary 4.4 (Negative answers to the dimension implications). *For every fixed $0 < \epsilon < 1/4$, none of (L), (P) and (Pn) holds universally with $\text{dc}(H)$ or $\text{dc}_{1/2}(H)$ on the left. For every fixed $C > 0$, choose*

$$0 < \epsilon < \min\{1/4, 1/(2C)\}, \quad \eta = C\epsilon.$$

The same three conclusions fail with $\text{dc}^{C\epsilon}(H)$ on the left, under each nondegenerate tie convention of Lemma 2.3. The counterexamples have proper deterministic, table-polynomial learners with $m = O_\epsilon(\log M)$ and $\tau = \Omega_\epsilon(1)$.

Proof. Fix the required constant accuracies and intersect the high-probability events of Theorems 4.1 and 4.2. Their failure probability is at most $2^{-I/2} + 2^{-(1-h_2(\eta))I/2} < 1$ for all sufficiently large N . Thus one matrix satisfies the two lower bounds simultaneously. Apply Lemma 4.3 for exact probabilistic dimension, and Theorems 5.1 and 5.2 for the proper deterministic learner. Since $M = N^3$ and $n = \log_2(2M)$, the dimension lower bounds grow exponentially in n , up to powers of $\log M$, whereas $m = O_\epsilon(n)$ and $1/\tau = O_\epsilon(1)$. No fixed polynomial in the proposed variables can dominate these lower bounds. $\square$

The strict-realization reduction in Lemma 2.3 gives the same lower bounds under either fixed tie rule and when every zero score counts as an error. Thus Corollary 4.4 covers all three nondegenerate conventions.

5 Efficient, proper and succinct learners

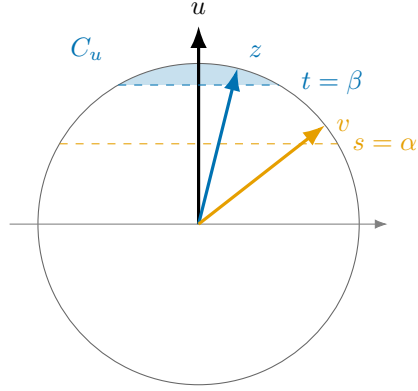
5.1 Cap mixtures and computation from an explicit table

The linear program in Lemma 3.3 has exponentially many possible rectangles. A fixed-dimensional template makes a different sampler available. Exact radial isotropy determines the geometric constant; a rational approximation and a finite direction net yield the finite-bit algorithm.

Algorithm 2: Cap sampler from a strict template

Input: Unpeeled points U , surviving rows V , and the uniform prior ν on V .

- 1 Use the fixed general-position perturbation and place V in radial-isotropic position in effective dimension j ;
 - 2 Set $\alpha = (2j)^{-1/2}$, $\beta = \sqrt{1 - \alpha^2}$; draw u uniformly on S^{j-1} ;
 - 3 Set $S \leftarrow U \cap \{x : \langle u, z_x \rangle \geq \beta\}$ and $T^\pm \leftarrow \{h : \pm \langle u, v_h \rangle > \alpha\}$;
 - 4 Choose the larger of T^+, T^- under ν ;
 - 5 **if** the positive side is chosen **then**
 - 6 **return** $(S, T^+, +1)$;
 - 7 The negative rectangle has a singleton side. Split the opposite side according to its labels in A , choose either color with probability $1/2$, and return that monochromatic rectangle;
-



$$\begin{aligned}
 t &= \langle u, z \rangle \geq \beta = \sqrt{1 - \alpha^2} \\
 s &= \langle u, v \rangle > \alpha \\
 \langle z, v \rangle &\geq ts - \sqrt{1 - t^2} \sqrt{1 - s^2} > 0 \\
 &\text{because } s^2 + t^2 > 1.
 \end{aligned}$$

The point cap and the row threshold differ. A negative row side is handled by a singleton-side split.

Figure 3: Cap geometry in a two-dimensional section. The drawing uses $\alpha = 1/2$ and $\beta = \sqrt{3}/2$. Blue denotes point vectors in the cap; the orange row vector satisfies the weaker row-side threshold. The strict inequality proves the common template sign used in Theorem 5.1.

Theorem 5.1 (Pointwise cap mixture and table-polynomial implementation). *Fix $d \geq 1$. Let F have a strict rank- d representation and no negative 2×2 , and let A be any monotone flip of its negative entries. Perturb the row vectors once into linear general position without changing F . At any nonempty survivor set, use the uniform row prior ν . There is a cap sampler, depending on the surviving rows but not on the unknown marginal, such that*

$$\forall x \in U, \quad \mathbb{E}[\mathbf{1}_S(x)\nu(T)] \geq \frac{c_d}{8d}, \quad c_d = \Pr_{u \sim \text{Unif}(S^{d-1})}[u_1 \geq \sqrt{1 - 1/(2d)}]. \quad (5.1)$$

Here $c_1 = 1/2$. Thus Theorem 3.1 applies with $\rho = 8d/c_d$ in the ideal real-arithmetic model.

For each fixed d , given the explicit table of A and rational template vectors of bit length at most b , there are randomized and deterministic implementations, polynomial in $|X|, |H|, b, 1/\epsilon$, for every $0 < \epsilon < 1/4$, with

$$m = O_d(\log |H| + \log(1/\epsilon)), \quad \tau = \Omega_d(\epsilon/\log(1/\epsilon)).$$

Queries are intersections of stored cap tests, or such tests split by one row label or one point's row labels. The running time is polynomial in the supplied table size.

Proof idea. Isotropy forces a constant amount of row mass beyond the two thresholds $\pm\alpha$. The cap geometry in figure 3 makes the selected product monochromatic in the template. A negative product has a singleton side, and random color splitting preserves half its expected coverage after flips. A rational transform and a finite direction net give the finite-bit implementation. Proof: see Appendix C.1.

Algorithm 3: Proper RECTIFY

```
1 Run RECTIFY at accuracy  $\epsilon/4$ , interrupting it at the first negative peel;
2 if no negative peel occurs then
3   return the all-positive member;
4 Use the rectangle's singleton point, or test its singleton row at loss threshold  $\epsilon/2$ ;
5 if the row passes then
6   return that row;
7 Otherwise locate the unique heavy negative target point  $p$  in the rectangle by binary search;
8 Query negative mass outside  $p$ ;
9 if the answer exceeds  $\epsilon/4$  then
10  | Binary-search the rows through  $p$  using negative-union queries; return the target;
11 else
12  | Test at most  $\lceil 8/\epsilon \rceil + 1$  rows negative at  $p$ ; return a row whose loss answer is at most  $\epsilon/2$ ;
```

For $d = 6$, $c_6 \approx 0.0003509774$ and $8d/c_d \approx 136760.93$. These numerical values describe the ideal sampler. The rational-net construction has worse, dimension-dependent constants but the same stated asymptotics. A low-rank template alone is insufficient: the no-negative- 2×2 condition is used exactly at the splitting step.

5.2 Proper learning without a dual-plane search

Every grid family contains the all-positive row: take intercept $b = 2N^2$. Proper learning therefore does not require testing random lines when the total negative mass is small. The first substantial negative rectangle supplies the missing localization information.

Theorem 5.2 (Proper RECTIFY). *For every finite nonempty class H and $0 < \epsilon < 1/4$, suppose H contains the all-positive hypothesis and has no negative 2×2 . Given a rectangle sampler of coverage r , it has a distribution-independent proper SQ learner of expected error at most ϵ , using*

$$m_{\text{proper}} \leq m_{\text{RECTIFY}}(\epsilon/4) + \lceil \log_2 |X| \rceil + \lceil \log_2 K \rceil + \lceil 8/\epsilon \rceil + 4 \quad (5.2)$$

queries at the tolerance of RECTIFY with accuracy $\epsilon/4$. For the incidence flips,

$$m_{\text{proper}} = O(\log M + \epsilon^{-1} + \log(1/\epsilon)), \quad \tau = \Omega(\epsilon/\log(1/\epsilon)).$$

The cap implementation is polynomial-time in the explicit table. Its finite-net deterministic variant is also proper and has the same asymptotic query bound, with constants depending on the template dimension.

Proof idea. At the first negative peel, a rejected singleton row intersects the target's negative support in one heavy point. The other rows through that point have disjoint negative supports away from it. A large remaining negative mass identifies the target; a small one makes one of a short list of tested rows accurate.

Proof: see Appendix C.2.

The all-light case requires no aspect-ratio argument. At the first negative peel, either the singleton candidate is already good, or its intersection with the target contains one heavy negative atom. The other members of the star at that atom have disjoint negative supports.

5.3 Succinct learning by order medians

The real-grid structure permits a learner that does not construct the truth table, its cap mixture, or a radial-isotropic transform. Its negative labeled measure is supported on one unknown affine line. Integer order, rather than pairwise sampling, identifies that line unless one negative point already carries enough mass for a star reduction. This resolves the all-light case with a quadratic, rather than linear, logarithmic query bound.

Write $\mu^-(E) = \Pr_D[p \in E, h^*(p) = -1]$, $\beta = \mu^-(X)$, and $\zeta = \epsilon/4096$. We analyze answers of error at most ζ . In a finite-bit implementation request raw SQ tolerance $\zeta/2$ and round each response with additional error at most $\zeta/2$, using $O(\log(1/\epsilon))$ bits after the binary point. This makes the algorithm a finite-bit computation without assuming exact real arithmetic. Points have coordinates (x, y) ; the label in an SQ is a separate variable.

Lemma 5.3 (Noisy integer medians). *Suppose $\beta > 0$, $|\hat{\beta} - \beta| \leq \zeta$, and $\beta \geq 512\zeta$. Let f be integer-valued in $[L, H]$ on the negative support. Binary search using the query $\mu^-\{f \leq t\}$ and threshold $\hat{\beta}/2$ returns $v \in [L, H]$ with*

$$\mu^-\{f \leq v\} \geq \beta/2 - 3\zeta/2, \quad \mu^-\{f < v\} < \beta/2 + 3\zeta/2.$$

If the answer $\hat{m}$ to $m = \mu^-\{f = v\}$ is at most $\hat{\beta}/8$, then

$$\beta/2 - 3\zeta/2 \leq \mu^-\{f \leq v\} < 5\beta/8 + 21\zeta/8. \quad (5.3)$$

Otherwise $m > \beta/9$.

Proof: see Appendix D.1.

For a point $p = (p_x, p_y)$, its template star contains the slopes

$$1 \leq a \leq t(p) := \min\{N, \lfloor (p_y - 1)/p_x \rfloor\}, \quad b = p_y - ap_x.$$

The upper intercept bound is automatic. Distinct star lines intersect only at p . Say that a flip matrix is *star-regular* if for every point with $t(p) \geq 64\lceil \log_2(2N^3) \rceil$, at least $t(p)/4$ of these rows are negative at p .

Theorem 5.4 (Succinct median learner). *Let $0 < \epsilon < 1/4$. Suppose entries of a real-grid incidence flip A can be evaluated in time $E(n)$, where $n = \log_2(2N^3)$. There is a deterministic improper distribution-independent SQ learner of error at most ϵ , using $O(n^2)$ queries at raw tolerance $\epsilon/8192$, with total computation polynomial in $(n, 1/\epsilon, E(n))$ and an efficiently evaluable output. If A is star-regular, there is a proper randomized learner with*

$$m = O(n^2 + 1/\epsilon + \log(1/\epsilon)), \quad \tau = \epsilon/8192,$$

with the same polynomial-time property and expected error at most ϵ against every valid oracle.

Proof idea. For a proposed slope a , the residual $y - ax$ increases along the target line when $a < a^*$ and decreases when $a > a^*$. Balanced prefix and residual-median sets are therefore nested or oppositely ordered. Their intersection masses differ by a constant fraction of the total negative mass, as illustrated in figure 4. Heavy atoms lead to the same star reduction as proper RECTIFY. Proof: see Appendix D.2.

Algorithm 4: Order-median learner

Input: An entry evaluator for A , the scale N , and $0 < \epsilon < 1/4$.

- 1 Set $\zeta = \epsilon/4096$. Query the total negative mass β ;
- 2 **if** $\hat{\beta} \leq \epsilon/4$ **then**
- 3 **return** the all-positive row;
- 4 Find a noisy median of the negative x -marginal;
- 5 **if** *its atom is heavy* **then**
- 6 **locate** its negative point and use the star completion;
- 7 Fix the balanced prefix P at the light median; maintain a slope interval containing a^* ;
- 8 **while** *the slope interval has more than one element* **do**
- 9 At its midpoint a , find a noisy median v of $f_a(x, y) = y - ax$;
- 10 **if** *the median atom is heavy* **then**
- 11 Its intersections with P and P^c either identify $(a^*, b^*) = (a, v)$ or locate one negative point for the star completion;
- 12 Query $\mu^-(P \cap \{f_a \leq v\})$;
- 13 Compare its answer to $3\hat{\beta}/8$ and retain the correct strict half of the slope interval;
- 14 Find the constant residual b^* and return h_{a^*, b^*} ;

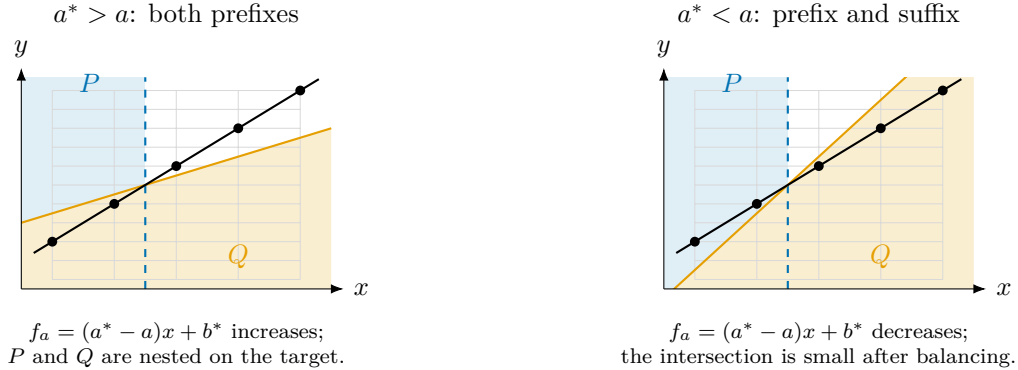


Figure 4: The order test behind Theorem 5.4. The target is $y = 2x + 1$. The blue region is an x -prefix; orange denotes $y - ax \leq v$. On the target, the orange region becomes a prefix when the trial slope is too small and a suffix when it is too large. The proof uses noisy median mass bounds to separate these cases by a constant fraction of the negative mass.

6 Conditional explicit families

6.1 Conditional succinctly computable separations

We now distinguish explicit evaluation from explicit deterministic selection of a good table. A seeded family is *succinctly evaluable* if its entries are computable in time polynomial in the example encoding length and the supplied seed length. The result below gives a negligible exceptional set of keys. It does not efficiently recognize good keys or specify an efficiently computable deterministic sequence of them.

Cryptographic assumption. Fix a constant $\delta > 0$, decreasing it to be less than one if necessary. Assume a polynomial-time bit-output pseudorandom function family $\{f_s\}_{s \in \{0,1\}^\sigma}$, defined on inputs long enough to encode a grid row and point, such that every oracle algorithm running in time at most 2^{σ^δ} has negligible distinguishing advantage between f_s for uniform s and a uniform random function. Negligibility is in σ . We use this assumption directly. The one-way-function, pseudorandom-

generator and pseudorandom-function constructions of Goldreich et al. [20], Håstad et al. [23] provide background for this assumption. The quantitative security hypothesis above is assumed directly; the lower bounds below do not require a separate quantitative reduction from one-way functions.

Take $n = 3b + 1$, $N = 2^b$, so that the domain size is exactly $2N^3 = 2^n$. Encode (a, b', x, y) injectively in $O(n)$ bits and pad if needed to the PRF input length. Define A_s to be positive off template incidences and to have sign $2f_s(a, b', x, y) - 1$ on incidences. Thus a bit one flips a negative entry to positive.

Lemma 6.1 (Small subgrid counting and decision cost). *At grid scale $k \geq 2$, put $K_k = 2k^3$ and*

$$I_k = 2k^4 - \left(\frac{k(k+1)}{2} \right)^2, \quad D_k = \left\lfloor \frac{k}{24 \log_2 k} \right\rfloor.$$

Under independent fair flips,

$$\Pr[\text{signrank}(A) \leq D_k] \leq 2^{-k^4/2}.$$

Whether $\text{signrank}(A) \leq D_k$ can be decided after reading at most $I_k \leq 2k^4$ oracle bits, in time $2^{O(k^3 D_k \log k)} = 2^{O(k^4)}$ when $D_k \geq 1$. When $D_k = 0$, the event is empty and is decided immediately.

Proof: see Appendix E.1.

Theorem 6.2 (Conditional almost-every-key explicit lower bounds). *Assume the stated PRF security for a fixed $\delta > 0$. Fix $c \geq 1$, $0 < \epsilon < 1/4$, $n = 3b + 1$, and put*

$$\sigma = n^{\lceil (4c+2)/\delta \rceil}.$$

For every key, entries of A_s and hypotheses in H_{A_s} are evaluable in $\text{poly}_c(n)$ time. Every key has RECTIFY and proper learners with $O_\epsilon(n)$ queries and tolerance $\Omega_\epsilon(1)$. For all sufficiently large admissible n , outside a negligible fraction of keys,

$$\boxed{\text{dc}(H_{A_s}) \geq \frac{n^c}{30 \log_2 n}.$$

Here the learner with $O(n)$ queries may use table-polynomial computation. A succinct polynomial-time implementation with $O(n^2)$ queries is established in Section 5.3.

Proof: see Appendix E.2.

For the linear implication choose any $c > 1$. For any proposed fixed-degree polynomial in $(m, 1/\tau, n)$ choose c larger than its degree, holding ϵ fixed. The lower bound then exceeds that polynomial for all sufficiently large n . The polynomial key-length exponent depends on c ; this statement is a hierarchy of succinctly evaluable families, not one polynomial-key family that beats every polynomial. Choosing instead $k = n^{\log_2 n}$ and $\sigma = \lceil k^{6/\delta} \rceil$ makes the same proof give superpolynomial dimension with quasipolynomial seed length and evaluation time. It does not settle polynomial-time deterministic explicit selection in the direction of Hatami et al. [24, Problem 4.1].

Proposition 6.3 (Unconditional polynomial advice). *Fix $c \geq 1$ and $0 < \epsilon < 1/4$. For all sufficiently large $n = 3b + 1$, there is a family of incidence-flip classes on 2^n points indexed by advice strings of length $O_c(n^{4c})$. Every entry is evaluable in $\text{poly}_c(n)$ time. For all but a $2^{-\Omega_c(n^{4c})}$ fraction of advice strings,*

$$\text{dc}(H) \geq \frac{n^c}{30 \log_2 n}.$$

Every advice string admits a proper deterministic learner running in $\text{poly}_{c,\epsilon}(n)$ time, with $O_{c,\epsilon}(\log n)$ queries at tolerance $\Omega_\epsilon(1)$.

The advice stores only the flip bits of a polynomial-size subgrid. Collapsing all other points to one positive point gives a polynomial-size effective table, so its proper cap learner has $O_{c,\epsilon}(\log n)$ queries. The full construction and counting proof appear in Appendix E.5.

This is an almost-every-advice result. Neither it nor Theorem 6.2 specifies a polynomial-time deterministic selector of good descriptions. The PRF result additionally fixes a cryptographic family and supports the hidden-key comparison in Proposition 8.3.

Lemma 6.4 (Regular stars for almost every key). *Use the star-regularity definition in Section 5.3: every template star with at least $64\lceil n \rceil$ rows has at least one quarter of its rows negative at its center. This property is independent of ϵ ; the algorithm tests it only for stars larger than $C_0 = \max\{64\lceil n \rceil, \lceil 128/\epsilon \rceil\}$. For independent fair incidence flips, the probability of failure of star-regularity is at most e^{-3n} , where $n = \log_2(2N^3)$. Under the PRF assumption and parameters of Theorem 6.2, failure occurs for only a negligible fraction of keys.*

Proof: see Appendix E.3.

Corollary 6.5 (Conditional polynomial-time, seed-known separation). *Under the PRF assumption, for every fixed $c \geq 1$ and $0 < \epsilon < 1/4$, all but a negligible fraction of keys in Theorem 6.2 define classes with*

$$\text{dc}(H_{A_s}) \geq \frac{n^c}{30 \log_2 n}, \quad m = O_\epsilon(n^2), \quad \tau = \Omega_\epsilon(1),$$

and a proper learner running in $\text{poly}_{c,\epsilon}(n)$ time given the key. Every key has the corresponding deterministic improper polynomial-time learner, independently of star-regularity.

Proof: see Appendix E.4.

Taking $c > 2$ refutes the linear query/tolerance implication even when the learner must run in time polynomial in n . Taking c larger than twice a proposed polynomial degree refutes each fixed-degree polynomial in $(m, 1/\tau, n)$ within this computational model. The key length and runtime exponents grow with c , so this does not refute a bound whose right side also charges an arbitrary polynomial of the actual runtime or expanded description size. Reducing the succinct query bound to $O(n)$ remains open.

7 Sufficient conditions and their limits

The rectangle condition admits a communication interpretation, but neither condition characterizes SQ learnability. Spectral certificates give a separate sufficient condition for hardness. Cube halfspaces and parities distinguish these directions.

7.1 Product-distribution communication

Let $\text{AC}_\times(H)$ be the supremum over product distributions $\mu \times \nu$ of the minimum expected bit cost of an exact deterministic communication protocol for $(x, h) \mapsto h(x)$. The protocol may depend on the two distributions. It must compute the function on every pair, including pairs of zero probability. Leaves of a binary deterministic protocol are monochromatic rectangles.

Theorem 7.1 (Communication as a sufficient condition). *For every finite class,*

$$\text{rect}(H) \geq 2^{-2\text{AC}_\times(H)-1}, \quad \text{AC}_\times(H) \leq 2 \text{rect}(H)^{-1}.$$

In particular, for every $0 < \epsilon < 1/4$, if $\text{AC}_\times(H) \leq c$, one distribution-independent SQ learner satisfies (2.1) using

$$m = O(4^c(\log |H| + \log(1/\epsilon))), \quad \tau = \Omega\left(\frac{\epsilon}{4^c \log(1/\epsilon)}\right).$$

Proof: see Appendix F.1.

The cube halfspaces in Theorem 7.9 show non-necessity here too. Under the uniform product, their largest rectangle has mass at most $e^{-n/4}$. The first proof therefore gives expected communication at least $n/(8 \ln 2) - 1/2$, even though they have a distribution-independent polynomial-query learner. This direct consequence needs no Gap-Hamming reduction. Bounded product-average communication and a rectangle ratio bounded below are equivalent up to the displayed nonlinear conversion; neither characterizes SQ learning.

7.2 Spectral certificates and why they miss this

Lemma 7.2 (Finite energy and threshold counting). *Let P_0 be a probability distribution on a finite set Z , and let $r_1, \dots, r_K \in L^2(P_0)$. Set*

$$\Gamma_{ij} = \mathbb{E}_{P_0} r_i r_j, \quad \Lambda = \|\Gamma\|_{\text{op}}.$$

For every $a : Z \rightarrow \mathbb{R}$,

$$\sum_{j=1}^K (\mathbb{E}_{P_0} r_j a)^2 \leq \Lambda \mathbb{E}_{P_0} a^2. \quad (7.1)$$

In particular, if $|a| \leq 1$ and $\tau > 0$, then

$$|\{j : |\mathbb{E}_{P_0} r_j a| > \tau\}| \tau^2 \leq \Lambda. \quad (7.2)$$

More generally, for any finite list of real numbers c_i , $\tau \geq 0$, and $\sum_i c_i^2 \leq \Lambda$, one has $|\{i : \tau < |c_i|\}| \tau^2 \leq \Lambda$.

Proof: see Appendix F.2.

Theorem 7.3 (Joint-distribution SQ barrier). *Let μ be a full-support distribution on X and let $P_0 = \mu \otimes \text{Unif}\{-1, +1\}$. For $j \in [K]$, let P_j be the labeled distribution arising from some D_j and $h_j \in \mathcal{H}$. Put*

$$r_j = \frac{P_j}{P_0} - 1, \quad \Gamma_{ij} = \mathbb{E}_{P_0} r_i r_j, \quad \Lambda = \|\Gamma\|_{\text{op}}, \quad R = K/\Lambda.$$

Any randomized adaptive (m, τ) -SQ learner satisfying (2.1) obeys

$$1 - 2\epsilon \leq \sqrt{\frac{\Lambda}{K}} + \frac{m\Lambda}{K\tau^2}, \quad \text{equivalently} \quad \frac{m}{\tau^2} \geq (1 - 2\epsilon)R - \sqrt{R}. \quad (7.3)$$

Consequently, for $\epsilon < 1/4$,

$$R \leq 4 + 4m/\tau^2. \quad (7.4)$$

Proof: see Appendix F.3.

The stated fixed-marginal barrier. Set $D_j = \mu$ for every j . Then $r_j(x, y) = y h_j(x)$ and

$$\Gamma_{ij} = G_{ij} = \mathbb{E}_{\mu} [h_i h_j].$$

For $R = K/\|\Gamma\|_{\text{op}} \geq 16$, (7.3) implies

$$m \geq \frac{1}{4} R \tau^2. \quad (7.5)$$

Indeed, $1 - 2\epsilon > 1/2$ and $\sqrt{R} \leq R/4$. The cutoff matters: a uniformly positive lower bound proportional to $R\tau^2$ cannot hold for all tiny families, since a singleton needs no queries. The next theorem supplies the accompanying dimension lower bound.

7.2.1 Spectral sign-rank bounds

We use the spectral method associated with Forster [18], but include its geometric step rather than treating the published theorem as a proof.

Theorem 7.4 (Spectral sign-rank bounds). *If $A \in \mathbb{R}^{N \times K}$ satisfies $|A_{xj}| \geq 1$ for all entries, then*

$$\text{signrank}(\text{sign } A) \geq \frac{\sqrt{NK}}{\|A\|_{\text{op}}}. \quad (7.6)$$

If $M \in \{-1, +1\}^{N \times K}$ and $G = M^\top \text{diag}(\mu)M$ for any distribution μ on its rows, then

$$\text{signrank}(M) \geq \sqrt{K/\|G\|_{\text{op}}}. \quad (7.7)$$

Proof: see Appendix F.4.

Theorems 7.3 and 7.4 prove the valid conditional assertion in the proposed spectral barrier: a certificate with ratio R yields dimension lower bound $\sqrt{R}$ and query lower bound $\Omega(R\tau^2)$ when R is large. They do not prove that all explicit sign-rank lower bounds admit such an ordinary label-Gram certificate. The following family disproves that extra assertion in precisely this form.

7.2.2 Projective planes

Theorem 7.5 (Projective-plane incidence classes). *For each prime $q \geq 2$, let X_q be the points of the projective plane over $\mathbb{F}_q$, and let $\mathcal{H}_q$ have one hypothesis per line, positive on that line and negative elsewhere. Set $N = q^2 + q + 1$ and $k = q + 1$. Then:*

$$\frac{q+1}{\sqrt{q}} \leq \text{dc}(\mathcal{H}_q) \leq 2q+3, \quad (7.8)$$

$$\frac{|J|}{\|[\mathbb{E}_\mu h_i h_j]_{i,j \in J}\|_{\text{op}}} \leq 16 \quad \text{for every distribution } \mu \text{ and nonempty } J \subseteq \mathcal{H}_q. \quad (7.9)$$

There is a distribution-independent N -query tolerance- τ learner of loss at most τ . For each fixed D and $0 < \epsilon < 1/4$, there is a learner allowed to depend on D , with tolerance $\epsilon/2$ and

$$m_D \leq 1 + s + \binom{s}{2}, \quad s = \left\lceil \frac{4 \log N + 2}{\epsilon} \right\rceil,$$

that has loss at most ϵ for every target. Here $\log$ is natural. Nevertheless every single learner satisfying (2.1), with $m \geq 1$, $0 < \tau \leq 1$, and $\epsilon < 1/4$, satisfies

$$\text{dc}(\mathcal{H}_q) \leq 19m/\tau^2. \quad (7.10)$$

Proof: see Appendix F.5.

Why projective planes do not separate. The family has unbounded dimension while all the proposed ordinary label-Gram ratios remain bounded. This is a genuine failure of the stated universality premise, not a counterexample to OQ2. The efficient learner needed for such a separation is impossible here: the target-dependent balanced marginals produce a joint-distribution ratio of order q . The lower-bound witness changes entry magnitudes, which the ordinary label Gram matrix cannot capture. This also verifies an explicit separation between fixed-distribution and distribution-independent SQ learning: $m_D = O(n^2/\epsilon^2)$ at tolerance $\epsilon/2$, whereas for $q \geq 16$ any common learner needs $m/\tau^2 \geq q/4$. Since $N \leq 2q^2$, the latter is at least $2^{(n-1)/2}/4$. The dimension lower bound is at least $2^{(n-1)/4}$. None of these fixed- D learners can be substituted for the single algorithm in OQ2.

7.2.3 Transcript compression

Theorem 7.6 (Marginal-branching and approximate probabilistic bounds). *Let $H \subseteq \{-1, +1\}^X$ be finite and nonempty, $m \geq 1$ an integer, $0 < \tau \leq 1$, and $0 < \epsilon < 1/4$. Put $B = \lceil 2/\tau \rceil + 1$. Suppose a depth-at-most- m deterministic (m, τ) -SQ learner satisfies (2.1) for this H and ϵ . Decompose each query as*

$$q(x, y) = a(x) + yb(x), \quad a(x) = \frac{1}{2}(q(x, +1) + q(x, -1)), \quad b(x) = \frac{1}{2}(q(x, +1) - q(x, -1)).$$

Construct its marginal-response tree as follows. If a is constant, retain only the answer equal to that constant. Otherwise retain the B equally spaced answers in $[-1, 1]$. Let u be the maximum number of nonconstant- a nodes on a retained root-to-leaf path. Then

$$\text{dc}(\mathcal{H}) \leq (m + 1)B^u. \quad (7.11)$$

In particular, a deterministic correlational-only learner has $\text{dc}(\mathcal{H}) \leq m + 1$. For an arbitrary randomized (m, τ) -SQ learner with worst-case query budget m satisfying (2.1) under the same hypotheses,

$$\text{dc}^\epsilon(\mathcal{H}) \leq B^m. \quad (7.12)$$

Proof: see Appendix F.6.

The deterministic argument does not derandomize the learner: fixing its seed need not preserve its expected guarantee. The probabilistic bound is exponential, not polynomial, and gives no exact $\text{dc}_{1/2}$ bound.

Proposition 7.7 (Affine-line quantitative comparison). *For the p^2 affine nonvertical lines on $\mathbb{F}_p^2$, the joint-distribution ratio in Theorem 7.3 can be chosen as*

$$R = \frac{2p}{1 + (p - 1)^{-2}} \geq p.$$

Thus every common learner of constant expected error $\epsilon < 1/4$ has $m/\tau^2 = \Omega(p)$. More generally, if on every instance it achieves correlation at least $2\gamma > 0$ with probability at least α , then

$$m \geq \tau^2 \left(\alpha R - \frac{1}{4\gamma^2} \right).$$

At Feldman's weak-learning parameters $\tau = 1/t$, $\gamma > 1/t$, $t = (p/32)^{1/4}$ and $\alpha = 2/3$, this gives $m = \Omega(\sqrt{p})$; at constant tolerance and constant advantage it gives $m = \Omega(p)$.

Proof: see Appendix F.7.

7.3 Rectangle geometry: tight examples and non-necessity

Theorem 7.8 (Global projective-plane rectangle ratio). *For the point-line incidence class of $\text{PG}(2, q)$, over any prime-power order $q \geq 2$ for which the plane is defined,*

$$\frac{1}{8(q+1)} \leq \text{rect}(H_q) \leq \frac{q}{(q+1)^2}.$$

Thus the infimum over all product distributions is $\Theta(1/q)$, not merely the maximum rectangle under the uniform product measure.

Proof: see Appendix F.8.

What is and is not tight. Theorem 7.3 and the balanced target-dependent marginals of Theorem 7.5 give $m/\tau^2 = \Omega(q) = \Omega(\rho)$ for this family. At a fixed positive tolerance this yields $m = \Omega(\rho)$. Theorem 3.1, however, uses $\tau = \Theta_\epsilon(1/\rho)$ and has $m/\tau^2 = O_\epsilon(\rho^3 \log K)$. Thus the geometry is sharp and a linear inverse-rectangle dependence is unavoidable at fixed tolerance, but the joint (m, τ) dependence of Theorem 3.1 is *not* established as tight up to logarithms. Lowering tolerance cannot be ignored in a tightness claim.

Theorem 7.9 (Small rectangle ratio does not prevent SQ learning). *For odd n , let*

$$X_n = \{-1, +1\}^n, \quad H_n = \{x \mapsto \text{sign}\langle w, x \rangle : w \in \{-1, +1\}^n\}.$$

Then $\text{dc}(H_n) \leq n$ and $\text{rect}(H_n) \leq e^{-n/4}$. For every $0 < \epsilon < 1/4$, one deterministic distribution-independent SQ learner satisfies (2.1) with $m = O(n^3/\epsilon^2)$ and $\tau = \epsilon/(8n)$.

Proof: see Appendix F.9.

For contrast, all parities on $\{-1, +1\}^n$ have an orthonormal label Gram matrix under the uniform marginal: the product of distinct characters is a nonconstant character, whose mean cancels by flipping one coordinate. Theorem 7.3 therefore gives $m/\tau^2 = \Omega(2^n)$ at any fixed accuracy below $1/4$. The same Walsh orthogonality bounds every uniform monochromatic rectangle mass by 2^{-n} . Small rectangles can therefore occur on either the easy or the hard side of SQ learning.

For an ordinary Gram matrix $G = \mathbb{E}_\mu h h^\top$ on K rows, let $R = K/\|G\|_{\text{op}}$. A monochromatic rectangle $S \times T$ satisfies

$$\mu(S)|T|^2 \leq \mathbb{E}_\mu \left(\sum_{h \in T} h(x) \right)^2 \leq \|G\|_{\text{op}}|T|.$$

Thus its mass under μ times the uniform row prior is at most $1/R$. A large ordinary certificate supplies both small rectangles and the SQ lower bound; small rectangles alone supply neither converse. This distinction is depicted in figure 5.

Why the spectral certificates miss the counterexample. The ordinary-Gram estimate above shows why large rectangles exclude a large fixed-marginal certificate. The balanced projective-plane certificate is a joint-distribution certificate, not one of these bounded ordinary-Gram ratios. For the flip family, Theorems 7.3 and 3.1 give $\mathfrak{J}(H_A) = O_\epsilon(\log M)$ while $\text{dc}(H_A) = \Omega(N/\log M)$. This is the promised failure of joint-certificate completeness.

Theorem 7.10 (Marginal adaptation, under a small-query budget). *Let H_A be a flip class at scale N with $M = N^3$, let $m \geq 1$ be an integer, and let $0 < \tau \leq 1$ and $0 < \epsilon < 1/4$. Every deterministic (m, τ) -SQ learner with worst-case query budget m satisfying (2.1) obeys*

$$u \geq \frac{\ln \text{dc}(H_A) - \ln(m+1)}{\ln(\lceil 2/\tau \rceil + 1)},$$

where u is the marginal-branching depth defined in Theorem 7.6. In particular, if $\text{dc}(H_A) \geq c_0 N/\log M$ for an absolute $c_0 > 0$, then for any fixed $C > 0$, $m \leq (\log M)^C$ and all sufficiently large N ,

$$u = \Omega_C \left(\frac{\log N}{\log(2 + 1/\tau)} \right).$$

At fixed positive tolerance, the $\Omega(\log N)$ requirement matches the $O_d(\log M)$ marginal-query upper bound of the deterministic finite-net learner, at fixed accuracy.

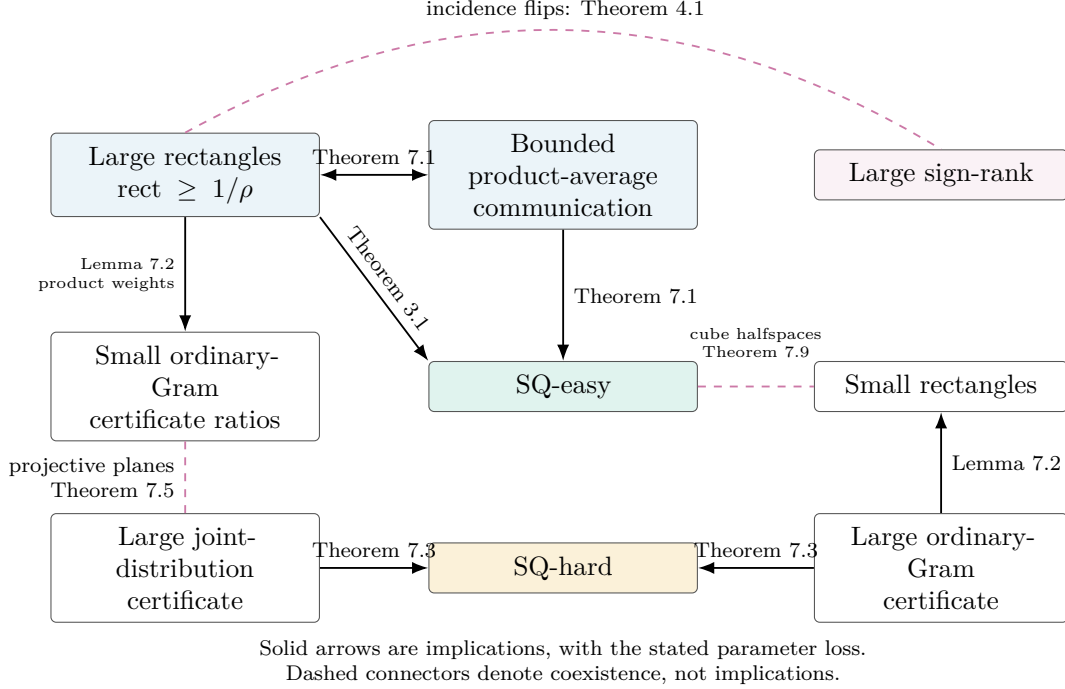


Figure 5: Which resources separate. Small rectangles alone do not imply SQ hardness: cube halfspaces are a counterexample. An ordinary spectral certificate separately forces small rectangles and SQ hardness. Projective planes require the stronger joint-distribution certificate, while incidence flips combine large rectangles with large sign-rank.

Proof: see Appendix F.10.

The query-budget qualification cannot be dropped: querying the correlation with every row is a deterministic correlational learner with $u = 0$ and $m = K$. Randomization is not intrinsically necessary for the separation, because the finite-net enumeration is deterministic. What is necessary for a small-query deterministic learner here is adaptation to marginal information. At inverse-polylogarithmic tolerance the proved lower bound loses a $\log \log M$ factor; no sharper matching statement is made.

7.4 Approximate deterministic embeddings

Corollary 7.11 (Published approximate rank as an SQ sufficient condition). *Fix $0 < \eta < 1/2$ and $0 < \epsilon < 1/4$. Suppose a total sign matrix has approximate sign-rank at most d in the deterministic-embedding sense of Bindu et al. [8]. Put $\bar{d} = \max\{2, d\}$ and*

$$\rho_\eta = \bar{d}^{C_\eta(\bar{d} + \bar{d}^2)},$$

where C_η is a sufficiently large constant valid in their rectangle theorem. Then

$$\text{rect}(H) \geq \rho_\eta^{-1}, \quad m = O(\rho_\eta(\log K + \log(1/\epsilon))), \quad \tau = \Omega\left(\frac{\epsilon}{\rho_\eta \log(1/\epsilon)}\right)$$

for one distribution-independent learner. Equivalently $\rho_\eta = \bar{d}^{O_\eta(\bar{d}^2)}$. The implicit constant depends only on η .

Proof: see Appendix F.11.

For comparison with randomized SQ dimension, see Appendix I. The construction of RECTIFY is independent of that characterization.

Class	Rectangle ratio	Dimension	Bounds on m/τ^2 and certificate
Incidence flips	$\geq 2^{-15}$	$\Omega(N/\log N)$ for most flips	$O_\epsilon(\log N)$ upper; counting proves high rank.
Projective planes	$\Theta(1/q)$	$[(q+1)/\sqrt{q}, 2q+3]$	$\Omega_\epsilon(q)$ lower; $O_\epsilon(q^3 \log q)$ rectangle upper; joint certificate.
Affine lines over $\mathbb{F}_p$	$\Theta(1/p)$	$\leq 2p+1$	$\Omega_\epsilon(p)$ lower; $O_\epsilon(p^2)$ exhaustive upper; joint certificate.
Cube halfspaces, odd n	$\leq e^{-n/4}$	$\leq n$	$O(n^5/\epsilon^4)$ upper via bounded-margin optimization.
All cube parities	$\leq 2^{-n}$	$\geq 2^{n/2}$	$\Omega_\epsilon(2^n)$ lower and $O_\epsilon(2^n)$ upper; orthogonal Gram matrix.

Table 4: Proved comparison bounds. Query lower bounds refer to fixed $\epsilon < 1/4$ and the adversarial SQ model. Entries give bounds established here, rather than a claim of globally optimal complexity. The affine rectangle estimate follows from the degree proof for Theorem 7.8 and the affine centered incidence norm.

Comparison of the example families. Table 4 collects the bounds. For affine lines, the degree argument in Appendix F.8 gives $\text{rect} \geq 1/(8p)$. The centered affine incidence matrix has norm $\sqrt{p}$ by Proposition 7.7; under uniform product weights the same bilinear argument gives rectangle mass at most $1/p$. Each affine row has p exceptional labels, so the moment-curve construction of Theorem 4.1 gives dimension at most $2p+1$. Exhaustively testing row correlations gives the displayed affine upper bound. For parities, the uniform Gram matrix is the identity and the spectral theorem gives dimension at least $2^{n/2}$; exhaustive testing matches the query lower bound at fixed accuracy.

8 Consequences for differentiable learning

A polynomial-time SQ learner can be simulated by gradient descent on an engineered differentiable model. The simulation charges its computation in the model size. This gives a consequence for the title question of Feldman et al. [17], while keeping the prescribed online-SGD problem separate.

8.1 An engineered gradient-descent corollary

The model used by Abbe et al. [2, Theorem 3d] is a directed acyclic neural network with $n+1$ inputs, including a constant input, one output, and one trainable weight per edge. Its activation is the piecewise-linear “two-stage ramp”

$$\sigma(z) = \begin{cases} -2, & z < -3, \\ z+1, & -3 \leq z \leq -1, \\ 0, & -1 < z < 0, \\ z, & 0 \leq z \leq 2, \\ 2, & z > 2. \end{cases}$$

Initialization is a specified polynomial-time computable map $W : \{0,1\}^r \rightarrow \mathbb{R}^p$, rather than an independent Gaussian law. Labels lie in $\{0,1\}$ and the loss is $\frac{1}{2}(f(x) - y)^2$. Full-batch updates reuse one iid sample S of size m_s , with a fixed constant step size and

$$\theta_{t+1} = \theta_t - \gamma g_t, \quad g_t \in \lambda \mathbb{Z}^p \cap [-1, 1]^p, \quad \left\| g_t - \frac{1}{m_s} \sum_{(x,y) \in S} [\nabla_{\theta} \frac{1}{2}(f_{\theta_t}(x) - y)^2]_1 \right\|_{\infty} \leq \frac{3\lambda}{4}. \quad (8.1)$$

Here $[\cdot]_1$ denotes coordinatewise clipping to $[-1, 1]$. The guarantee is $\mathbb{E}_{S,R} \sup_{(g_t) \text{ admissible}} L(f_{\theta_T})$: expectation over the reused sample and initialization follows the supremum over all admissible

rounded-gradient sequences. The supremum may depend on that entire sample. The constructed trajectories are differentiable and have gradient coordinates bounded by one, so clipping leaves them unchanged.

The imported simulation states that an SQ algorithm with k queries, tolerance τ_0 , r random bits and running time TIME has such a simulation with

$$\lambda = \tau_0/16, \quad T' = 2k, \quad m_s \lambda^2 > C(k \log(1/\lambda) + \log(1/\delta)), \quad p = \text{poly}(k, 1/\tau_0, r, \text{TIME}, 1/\delta), \quad (8.2)$$

whose expected square loss exceeds the SQ algorithm's by at most δ . The constant C is universal. These are the parameters of the full-batch part of that theorem; the sample is not refreshed at each step.

Corollary 8.1 (Engineered differentiable learning with high dimension). *Assume the PRF hypothesis of Theorem 6.2. Fix $c \geq 1$ and $0 < \epsilon < 1/4$. For every sufficiently large admissible encoding length n and every key s , the class H_{A_s} , encoded on $\{0, 1\}^n$, has an engineered neural network and initialization of the form above, each describable and evaluable in $\text{poly}_{c,\epsilon}(n)$ time, with $\text{poly}_{c,\epsilon}(n)$ parameters. After $O(n^2)$ full-batch updates using $\text{poly}_{c,\epsilon}(n)$ examples and the finite-precision gradient rule (8.1), thresholding the final output at $1/2$ has expected classification error at most ϵ for every marginal and every target in H_{A_s} . For all but a negligible fraction of keys,*

$$\text{dc}(H_{A_s}) \geq \frac{n^c}{30 \log_2 n}.$$

The model and initialization may depend on s , but are fixed before the marginal and target.

Proof. Use the deterministic improper order-median learner of Theorem 5.4 at accuracy $\epsilon/8$. It works for every key with $k = O(n^2)$ queries, tolerance $\Omega(\epsilon)$, and polynomial running time including query evaluation. Recode its labels and predictions from $\{-1, +1\}$ to $\{0, 1\}$. Its square loss is half its classification loss, hence at most $\epsilon/16$ against every valid response sequence. A dyadic tolerance τ_0 within a factor two of the advertised one preserves this guarantee and the asymptotic bounds.

Give the deterministic source learner one ignored random bit and apply (8.2) with $r = 1$ and $\delta = \epsilon/16$. Its initialization map may ignore this bit as well. The sample size and model size are polynomial in n for fixed c, ϵ , and the expected final square loss is at most $\epsilon/8$. A thresholding mistake implies $|f(x) - y| \geq 1/2$ and therefore square loss at least $1/8$. The expected classification loss is at most ϵ . The PRF sign-rank conclusion is Theorem 6.2; it is independent of the simulation. The deterministic source learner satisfies the simulation's supremum-over-responses premise without exchanging a supremum and an expectation. $\square$

The activation, architecture and initialization are engineered, and model size charges the simulated computation. In particular, the polynomial degree in the size bound can depend on c . The corollary establishes gradient-based distribution-independent learning of the class; it does not assert a superpolynomial separation between dimension and the size of this particular network.

8.2 The prescribed online-SGD question

For comparison, the remaining question fixes $X = \{-1, +1\}^n$ and a fully connected bias-free ReLU network

$$f_\theta(x) = \theta_L \sigma(\theta_{L-1} \sigma(\cdots \sigma(\theta_1 x))), \quad S = \sum_{i=1}^L n_i n_{i-1}.$$

Here $n_0 = n$ and $n_L = 1$. Each layer- i weight is initialized independently with law $N(0, 1/n_{i-1})$. With one fixed step size η , for $t = 0, \dots, T-1$ draw $x^{(t)} \sim D$ independently and set

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta} \ell(h^*(x^{(t)}) f_{\theta^{(t)}}(x^{(t)})), \quad \ell(z) = \log(1 + e^{-z}).$$

The returned predictor is

$$\hat{h}(x) = \text{sign} \left(\sum_{t=\lceil T/2 \rceil}^T f_{\theta(t)}(x) \right).$$

The architecture, step size and horizon are fixed for the class before D and h^* . Open Question 1 of Feldman et al. [17, Section 2] asks whether a guarantee $\mathbb{E} L_{D, h^*}(\hat{h}) \leq \epsilon < 1/4$ for every D, h^* , with expectation over both initialization and online sampling, forces $\text{dc}(H) \leq CTS$. Their Section 4 also asks for a polynomial bound in (T, S) . This question remains open here.

A population-gradient method with S coordinates, T calls, per-example coordinate bound B_g and adversarial Euclidean tolerance δ has an SQ simulation with

$$m = TS, \quad \tau = \delta / (B_g \sqrt{S}).$$

Query each gradient coordinate divided by B_g . After rescaling, each error is at most $\delta / \sqrt{S}$, so the total squared error is at most δ^2 . Induction over the calls proves the simulation. Conversely, the adaptive objective $J_q(t; x, y) = tq(x, y)$ has derivative $\mathbb{E} q$. These observations concern gradient-oracle models. A sampled online gradient need not lie in a prescribed small ball about the population gradient, and its sampling-law guarantee does not imply success against every perturbation in that ball; see also Abbe et al. [2], Feldman et al. [16].

Proposition 8.2 (A bias-free ReLU representation). *Pad the grid domain to a cube of dimension $n = \lceil \log_2(2N^3) \rceil$, giving every padding point label +1. Every member is represented exactly by a bias-free ReLU network with widths $(n, 2n, N+1, 1)$ and at most*

$$S = 2n^2 + 2n(N+1) + (N+1)$$

weights. Its common dimension complexity is at most $2N+1$. This proposition is an upper bound, not a claim that S parameters are necessary.

Proof: see Appendix G.1.

A fixed b -bit-per-weight architecture has at most 2^{bS} parameter choices. Representing all 2^N independent labelings of a fixed line in such an architecture would require $bS \geq N$. Unrestricted real weights have no fixed bit budget, so this finite-precision count does not give a lower bound for every real-weight representation of an individual row.

8.3 Hidden keys and class information

Fix the full-length template line $\ell_0 : y = x + 1$ and its uniform point distribution. Proposition 10.3 proves a sample lower bound for a single learner independent of a random table, even when that learner is told ℓ_0 and the marginal. The missing information is the unobserved label pattern. The computational analogue concerns a hidden PRF key.

Proposition 8.3 (Hidden-key computational hardness). *Under the PRF assumption of Theorem 6.2, no probabilistic polynomial-time learner independent of s , with a polynomial-time evaluable output and only polynomially many labeled samples from the uniform distribution on ℓ_0 , has average error at most $1/2 - \gamma(n)$ for uniform s , where $\gamma(n) \geq 1/\text{poly}(n)$. More precisely, a learner with $q(n)$ samples and advantage $\gamma(n)$ gives a PRF distinguisher of advantage at least*

$$\gamma(n) - \frac{q(n)}{2N}.$$

Proof: see Appendix G.2.

A guarantee for every key, or for all but a negligible fraction of keys, implies the average guarantee ruled out by Proposition 8.3. The original online-SGD formulation permits architecture, step size and horizon to depend on the class. Gaussian initialization alone does not remove that dependence. Thus the hidden-key reduction applies to key-independent finite-time procedures, rather than to every class-dependent design allowed in Open Question 1. A negative answer to that question requires a small compatible network and an analysis of the specified dynamics under every marginal.

9 Related work

SQ learning and dimension. Statistical-query learning accesses bounded expectations rather than individual examples [28]. Feldman’s randomized search characterization supplies a general dimension of this access model [15]; our use of it is confined to Appendix I. The question of Feldman et al. [17] asks whether one learner serving every marginal must admit a common low-dimensional linear representation. Our rectangle learner answers negatively without invoking the characterization. Euclidean representation complexity originates in Ben-David et al. [7]; probabilistic and approximate versions were developed by Chornomaz et al. [10] and Kamath et al. [26]. The exact and averaged variants require the distinct reductions in Theorem 4.2 and Lemma 4.3. In particular, uniform approximation on the sparse random matrix would not prove the averaged-dimension lower bound.

Correlational queries and the converse direction. The source note records a margin-complexity characterization of distribution-independent weak correlational learning [17, Section 3], citing Feldman [13]. Strong correlational learning is more restrictive: Feldman [14] proves hardness for conjunctions at vanishing target error. These two accuracy regimes are distinct. General SQ halfspace algorithms use rescaling and marginal information [6, 12]. The rescaling simulation depends polynomially on dimension, inverse accuracy and the logarithm of the inverse margin; finite-grid margin bounds control the latter through coordinate bit complexity [21]. The negative result here concerns reversing the implication without requiring a low-dimensional or bounded-margin representation.

Prior random-feature bounds. Karchmer and Malach [27] study approximate dimension complexity, random features and distribution-free differentiable learning. The source note states that their claimed upper bounds do not hold, attributing the correction to a 2026 personal communication from those authors [17, Section 3]. The probabilistic statement in Corollary 4.4 fixes one embedding law before both the target and marginal. Under these quantifiers it rules out an unrestricted polynomial bound from common SQ learnability to the nondegenerate averaged dimension. This comparison is about the precisely stated implication, rather than an alternative marginal-dependent random-feature distribution.

Sign-rank, margin and geometry. Forster’s normalization method provides spectral lower bounds [18]. The construction and the separation between sign-rank, VC dimension, average margin and rectangle ratio are due to Hatami et al. [24], building on Alon et al. [5]. We import their grid construction and Warren’s general strict-pattern theorem [33]; we derive the application-specific counting and probabilistic lower bounds. The homogeneous-rectangle theorem of Alon et al. [4] is used only to sharpen the inverse rectangle constant to 2^{15} . Our cap proof supplies an independent dimension-dependent constant. The minimax rectangle-mixture idea occurs in the average-margin argument of Hatami et al. [24, Theorem 3.1]; the new step here is its adversarial-oracle peeling/elimination analysis and bounded query budget. Approximate Forster transforms are prior algorithmic work [11]; Appendix J gives the fixed-dimensional implementation needed here.

The finite-field separation of 2017. Feldman [15, Theorem 7.7] studies affine lines on $\mathbb{F}_p^2$ and proves a distribution-independent weak-learning lower bound of $t/2 - 1$ queries at tolerance $1/t$, where $t = (p/32)^{1/4}$, for the stated accuracy and success-probability regime. Theorem 7.8 there gives a separately fixed-distribution learner with $O(1/\epsilon^2)$ queries of tolerance $\epsilon^2/13$. For the affine matrix, Proposition 7.7 applies the joint certificate to his affine class. At constant expected error it gives $m/\tau^2 = \Omega(p)$, hence $m = \Omega(p)$ at constant tolerance. The proposition also supplies the exact success-probability comparison at his tolerance. These are quantitative refinements of the hardness side; his class is not a small-complexity distribution-independent learner and therefore is not the present counterexample. Our real-grid monotone flips lie on the easy side of that universal-marginal requirement.

Approximate sign-rank, May 2026. Bindua et al. [8, Theorem 1.3] extend the large-rectangle property to approximate sign-rank and partial matrices. Their approximate rank fixes one *deterministic* column embedding and, for each row and each marginal, permits a new separating vector with error at most η . It is not the random-embedding quantity dc^η used here. Their theorem supplies row and column fractions $d^{-C_\eta d}$ and $d^{-C_\eta d^2}$, respectively, with C_η stated as $O(\log(1/(1 - 2\eta)))$, for fixed $0 < \eta < 1/2$; one may enlarge the constant and use $d \geq 2$. Corollary 7.11 converts this published geometric theorem into a common SQ learner with the displayed parameters. For partial matrices their rectangle is homogeneous on nonstar entries, possibly containing stars; we do not silently regard it as a full monochromatic rectangle of an arbitrary completion.

Index and list replicability, June 2026. Blondal et al. [9] prove $\text{Ind}_{\mathbb{Z}_2}(H) \leq 2 \text{LR}(H) - 1$ and show that projective-plane incidence classes have list replicability at most three and index at most five despite unbounded sign-rank. Our joint-distribution theorem shows that those same projective-plane classes are SQ-hard when a common learner must serve every marginal. Thus bounded list replicability or bounded index is not the missing SQ upper-bound criterion. Their separation concerns different parameters and does not resolve the question answered here. Earlier and concurrent topological bounds [19, 25], and the total Hamming-distance upper bounds of Göös et al. [22], likewise do not give the sought efficient explicit high-sign-rank selection for the constant-rectangle flip family.

The August 2026 conditional SQ theorem. A further comparison is Zhang et al. [34, Assumption 4.42 and Theorem 4.9]. Its SQ result assumes that the span of all seed-averaged terminal response functions, over complete response rules, already has dimension at most $B(1 + m/\tau^2)^k$. Under that additional assumption it proves a dimension upper bound. The extra response-rank hypothesis is not a consequence of the SQ interface proved there. Our counterexample rules out deriving such a universal polynomial hypothesis from distribution-independent SQ learnability alone. For the deterministic learners in Theorem 5.1, the universal mean-response-rank hypothesis would imply a polynomial dimension upper bound and hence contradict Theorem 4.1. Thus it cannot follow from the common SQ guarantee alone.

Learning versus communication and differentiable learning. The converse communication construction in Theorem 7.1 is Hatami et al. [24, Proposition 3.14]; its composition with RECTIFY yields the SQ sufficient condition and the halfspace boundary case. The proper completion and the order-median succinct learner exploit the real incidence geometry rather than invoking a generic protocol-to-learning conversion. The differentiable-learning simulations of Abbe et al. [2] use engineered models and initialization and charge the simulated computation in their network-size bounds. They do not preserve the remaining prescribed benign-SGD algorithm. The hidden-key bounds in Section 8.3 explicitly distinguish uniform key-independent learners from class-dependent architecture choices.

Neural dynamics and staircase structure. Malach et al. [30] compare gradient learning with tangent kernels under specified model conditions. The staircase analyses of Abbe et al. [1] and Abbe et al. [3] identify hierarchical structures learnable by particular regular-network dynamics under their distributional assumptions. Their conclusions concern those dynamics and data regimes. Corollary 8.1 instead composes a common SQ learner with the full-batch simulation of Abbe et al. [2]; its engineered architecture and initialization retain the simulation’s computational cost. The prescribed Gaussian-initialized online-SGD question therefore remains a separate problem.

10 Discussion and open problems

10.1 Distribution adaptation and simultaneous representation

Distribution independence constrains the learner to use one algorithm, rather than one algorithm per marginal. It does not impose a common feature map on its intermediate computations. Theorem 3.1 locates marginal mass and eliminates hypotheses sequentially; Theorems 4.1 and 4.2 show that realizing all signs simultaneously can require exponentially larger dimension. Rectangle structure makes the sequential task cheap without bounding the common representation dimension.

The separation uses the difference between general and correlational SQ access. Theorem 7.6 gives $\text{dc}(H) \leq m + 1$ for deterministic correlational learners. For the incidence flips, Theorem 7.10 forces $\Omega(\log N)$ marginal-dependent branchings at fixed tolerance under a polylogarithmic query budget, and the deterministic cap learner uses $O(\log M)$ stages. The claim has a query-budget hypothesis: exhaustive correlation testing uses no marginal-dependent branchings but makes one query per row. This is consistent with the margin-based characterization of distribution-independent weak correlational learning [13, 14].

An ordinary label-Gram certificate produces both a small-rectangle witness and an SQ lower bound. The constant-rectangle incidence flips exclude a large ordinary certificate, while counting still forces high sign-rank. The joint-distribution certificate is also polynomially bounded for these classes by Theorem 7.3 and their efficient SQ learner. The next proposition locates the limitation of source length; the following certificate proposition locates the limitation of the classical parameters.

Theorem 10.1 (Source-description length alone does not repair the implication). *Fix $0 < \epsilon < 1/4$. Interpret ℓ as the bit length of a finite learner program, including literal hard-coded constants and data, but do not charge running time or generated workspace. There is a computable sequence of incidence-flip classes and learners with*

$$n, \ell, m = O(\log N), \quad 1/\tau = O_\epsilon(1), \quad \text{dc}(H) = \Omega(N/\log N).$$

Consequently no universal bound $\text{dc}(H) \leq \text{poly}(m, 1/\tau, n, \ell)$ holds in this interpretation.

Proof: see Appendix H.1.

This does not produce an efficient explicit counterexample. It implies that a resource-aware repair must charge computation, generated memory, or an expanded circuit description, not just the length of the program that generates them. One genuinely stronger question is whether charging total preprocessing and query-evaluation time t yields a bound polynomial in $(m, 1/\tau, n, \ell, t)$ in a specified finite-bit model. This finite-bit resource question remains open.

Proposition 10.2 (The classical certificate barrier). *For the orientation in which rows are hypotheses, let*

$$m_{\text{avg}}(A) = \inf_{\mu \in \Delta(X)} \sup_{\text{strict unit-vector representations of } A} \min_h \mathbb{E}_{x \sim \mu} |\langle v_h, u_x \rangle|.$$

Then

$$\sqrt{\text{VC}(H)} \leq m_{\text{avg}}(A)^{-1} \leq \text{rect}(A)^{-1}.$$

In particular, if $\text{rect}(A)^{-1} \leq \text{poly}(n)$, these three classical lower-bound parameters cannot certify superpolynomial sign-rank. The bound concerns the three displayed certificates; actual sign-rank can be larger.

Proof: see Appendix H.2.

The chain and its original average-margin interpretation are due to Hatami et al. [24]; the proof is included to fix the orientation and strict-sign issue. A counterexample obtained with a polynomial inverse rectangle bound cannot be certified as high-sign-rank by those numerical parameters. Counting is already a way around that barrier, as Theorem 4.1 demonstrates. Therefore the barrier does not rescue the program-length conjecture. For an *efficiently explicit* sequence it motivates seeking a method beyond those parameters, in the direction of Hatami et al. [24, Problem 4.1].

The recent topological methods provide a distinct parameter, not one proved bounded by this chain. The checked results do not supply the missing efficient total construction: Frick et al. [19] prove near- $2k$ bounds for certain *partial* Gap Hamming matrices, while Göös et al. [22] bound the sign-rank of total k -Hamming-distance matrices by $2^{O(k)}$ independently of the bit dimension. These facts neither rule out a future topological route nor prove that it evades the rectangle obstruction on the particular flip family. The efficiently explicit total construction remains open.

Proposition 10.3 (Unseen independent labels). *Fix the full-length template line $\ell_0 : y = x + 1$ and the uniform distribution on its N points. Choose A by independent fair incidence flips. Let a single learner, chosen independently of A , receive T independent labeled samples from (D, h_{ℓ_0}) and no further information about A . Then*

$$\mathbb{E}_{A, \text{sample}, \text{coins}} L_{D, h_{\ell_0}}(\hat{h}) \geq \frac{1}{2}(1 - 1/N)^T \geq \frac{1}{2}(1 - T/N).$$

Thus a guarantee of average error at most ϵ , or a guarantee for every flip table, requires $T \geq (1 - 2\epsilon)N$. If $\epsilon < 1/4$ and $S \geq 2$, such a uniformly successful S -parameter learner satisfies $TS \geq \text{dc}(H_A)/3$ for every member of the flip family.

Proof: see Appendix H.3.

The final inequality in Proposition 10.3 requires one table-independent learner that succeeds on average over tables or uniformly over all tables. It fails as a sample requirement for an individual table: the all-positive table needs no samples. Therefore counting-based random tables obstruct hidden-table learning, but do not prove that every class-dependent benign-SGD choice satisfies the proposed dimension bound. A negative answer to Open Question 1 must exhibit a class of large dimension together with a small fully connected bias-free network and a proof that its specified Gaussian-initialized logistic-loss online updates learn under every marginal. For an efficiently explicit constant-rectangle candidate, Proposition 10.2 identifies which classical lower-bound certificates cannot establish the required dimension.

10.2 What a characterization must distinguish

Theorem 7.1 converts bounded product-average deterministic communication into a common SQ learner. Theorem 7.9 shows that neither bounded communication nor a large rectangle ratio is necessary. Cube halfspaces and parities both have exponentially small rectangles; only parities have the corresponding exponential SQ lower bound from orthogonality. A characterization therefore needs a representation or optimization resource beyond the rectangle ratio alone.

The standard rescaling route in the opposite direction simulates halfspace learning by SQ using dimension, accuracy and the logarithm of the inverse margin [6, 12]. Grid-coordinate margin bounds make the bit complexity of a representation relevant [21]. By contrast, the domain-size consequence

of Theorem 3.1 is direct:

$$m = O\left(\rho^2 \log(e|X|) + \rho \log(1/\epsilon)\right), \quad \tau = \Omega\left(\frac{\epsilon}{\rho \log(1/\epsilon)}\right).$$

It uses the combinatorial bound on VC dimension from rectangles, rather than a small-dimensional realization of the class.

Open problems.

1. Find one efficiently entry-computable family with polynomial-length descriptions that has constant or polynomial inverse rectangle ratio and dimension exceeding every polynomial in the encoding length. The PRF and advice constructions give one family for each prescribed degree; they do not select such a single polynomial-description family. This follows the explicit-construction direction of Hatami et al. [24, Problem 4.1].
2. Resolve the class-dependent benign online-SGD question stated in Section 8. Separately, determine whether charging preprocessing, query evaluation, generated memory and expanded model description yields a dimension bound in a specified finite-bit resource model.
3. Characterize distribution-independent SQ learnability by a resource that separates cube halfspaces from parities while including the rectangle condition as a sufficient case.
4. Determine the optimal joint query/tolerance tradeoff. The projective planes have $\rho = \Theta(q)$ and $m/\tau^2 = \Omega(\rho)$, while the rectangle theorem gives $O_\epsilon(\rho^3 \log |H|)$. Reducing the succinct median learner from $O(n^2)$ to $O(n)$ queries is a separate concrete target.

Scope. The mathematical statements concern the explicitly defined finite SQ model and nondegenerate classification loss. The conditional statements require the stated PRF security. Efficient evaluation from a seed or advice is distinguished from efficient deterministic selection of a good seed. The representation and simulation theorems retain their precision, initialization and computation hypotheses. Experimental scope is specified in Appendix M.

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A Oracle endpoints, conventions and constants

The inequalities below prove Theorem 3.1 for arbitrary valid oracle responses. In the table below $u = D(U)$, $s = D(S)$, $a = s/u$, $b = \nu(T)$, $w = D(S \cap \{h^* \neq c\})$, and hats denote oracle answers. “Pathwise” means that no random-oracle assumption is used. The conditioning for mixture expectations includes the returned $\hat{u}$ and all earlier history, but precedes the fresh rectangle draw.

	Inequality or invariant	Adverse valid answer or justification	Randomness used
R1	$\tau \leq \epsilon/8, \tau \leq r\epsilon/64$	Direct substitution of P ; no reply.	None
R2	Nonstop $u > \epsilon/4 - \tau \geq \epsilon/8$	$\hat{u} = u + \tau$ can keep the smallest u active.	Pathwise
R3	$\tau/u \leq r/8$	R1–R2.	None
R4	Acceptance gives $s \geq ru/2 - (1 + r/2)\tau$	$\hat{s} = s + \tau, \hat{u} = u - \tau$ favor accepting a small set.	Pathwise
R5	Accepted $a \geq 5r/16 \geq r/4$	Substitute R3 into R4.	None
R6	$a \geq 3r/4$ gives $\hat{s} - r\hat{u}/2 \geq ru/16 > 0$	$\hat{s} = s - \tau, \hat{u} = u + \tau$ oppose acceptance.	Pathwise
R7	Target in T implies $w = 0, \hat{w} \leq \tau < 2\tau$	Maximal reply $\hat{w} = w + \tau$.	Pathwise
R8	$\sum_{\text{elim}} b \leq \sum_{\text{elim}} -\ln(1 - b) \leq \ln K$	Target retained; normalized target weight grows from $1/K$ to at most one.	Pathwise
R9	Before peel j : $\epsilon/8 \leq (1 - r/4)^{j-1} \leq e^{-(j-1)r/4}$	R2, R5, and $\ln(1 - z) \leq -z$.	Pathwise
R10	Number of peels at most P	Take logarithms in R9.	None
R11	A peel has $w \leq 3\tau$; total wrong mass $\leq 3P\tau = \epsilon/8$	$\hat{w} = w - \tau$ permits the largest wrong mass. Peeled sets are disjoint.	Pathwise
R12	Normal stop: $u \leq \epsilon/4 + \tau$; loss $\leq \epsilon/2$	$\hat{u} = u - \tau$ permits the largest residual mass. Empty residual has mass zero.	Pathwise
R13	On acceptance, $G = a$ or b , so $G \geq ab$	Both admissible actions obey this, irrespective of the mismatch reply.	Pathwise
R14	$\mathbb{E}[ab \mid \text{past}] \geq r$	Integrate pointwise mixture coverage against D conditioned on U .	Fresh learner draw
R15	$\mathbb{E}G \geq r - (3r/4)\mathbb{E}b \geq r/4$	Rejection is confined to $a < 3r/4$ by R6; $b \leq 1$.	Fresh learner draw
R16	$\sum G \leq \ln K + \ln(8/\epsilon) + 1 = B$ before budget termination	R8; peel ratios telescope excluding the last, whose $a \leq 1$. No logarithm of zero.	Pathwise
R17	$e^{-g} \leq 1 - (1 - e^{-1})g$ for $0 \leq g \leq 1$	Convexity chord. After normal stopping put $G = 1$.	None
R18	$\mathbb{E}[e^{-G} \mid \text{past}] \leq e^{-r/8}; \mathbb{E}e^{-\sum G} \leq e^{-rR/8}$	R15, R17 and conditional expectation induction.	Learner draws; oracle arbitrary
R19	$\Pr[\text{budget failure}] \leq e^{B-rR/8} \leq \epsilon/2$	R16 and Markov’s inequality.	Learner draws; oracle arbitrary
R20	$\mathbb{E}L \leq \epsilon/2 + \Pr[\text{budget failure}] \leq \epsilon$	R12 and $0 \leq L \leq 1$.	Learner draws; oracle arbitrary

A randomized oracle can be included by conditioning on its past private coins; each displayed response bound still holds for every permitted answer. No step averages away an adversarial error.

The drift proof allows the oracle to choose either permissible action after seeing the current query. It does not give an oracle access to future learner randomness.

Proof of Lemma 2.3 and the zero-incidence convention. There are exactly $N(N+1)/2$ empty template rows: row (a, b) has no incidence precisely when $b > 2N^2 - a$. They are all-positive rows of every flip. Their weights k_h/I in Theorem 4.2 are zero; no uniform distribution on an empty set is formed. For nonempty rows these weights sum to one. For a fixed embedding, loss minimization ranges over finitely many attained sign patterns, so the infimum is a minimum. The weighted incidence mistake count is an integer divided by I ; some embedding has value at most its expectation bound. The same embedding is used for every row, with row-specific weights, exactly as required by the definition.

The working convention is $\text{sign}(0) = +1$. A positive row-specific bias in one common added coordinate turns zero scores positive without changing any negative score. Thus the strict rank in the counting argument is $d+1$, and the event is $\text{dc}^\eta < s_\eta$, not $\text{dc}^\eta \leq s_\eta$. If the sign convention is $\text{sign}(0) = -1$, use a negative bias. If zero predictions are counted as errors, replacing them by $+1$ can only reduce the number of mistakes, so the same lower bound still follows. Exact probabilistic dimension uses a full-support marginal; zero loss means that every point is correct. These observations make the conclusions insensitive to either fixed tie rule, without weakening the universal quantifiers over D and h . $\square$

B The grid construction and counting bound

Proof of Theorem 4.1. Define

$$u_p = (x^2, xy, y^2, x, y, 1), \quad v_\ell = (a^2, -2a, 1, 2ab, -2b, b^2 - 1/2).$$

Their dot product is $(ax + b - y)^2 - 1/2$. It is negative on incidences and positive on nonincidences because all coordinates are integers. Thus the six-dimensional strict factorization proves $\text{signrank}(F) \leq 6$. Both vectors are nonzero and may be normalized to unit vectors. Every row of A has at most N negative points. Enumerate all domain points by $t \in [K]$ and use $(1, t, \dots, t^{2N})$. For a row with negative index set S , the polynomial $\prod_{a \in S} (t - a)^2 - 1/2$ is negative exactly on S and positive at every other integer domain point. The empty product is one. This proves $\text{dc}(H_A) \leq 2N + 1$ for every member.

For fixed a, x , exactly $2N^2 - ax$ values of $b \in [2N^2]$ give an incidence with $y \in [2N^2]$. Summing proves (4.1). The lower bound uses $(N+1)/2 \leq N$. If two distinct affine lines share points (x, y) and (x', y') , subtraction gives $(a - c)(x - x') = 0$. If $a = c$, their shared point forces equal intercepts, a contradiction. Otherwise $x = x'$ and then $y = y'$. Hence F , and therefore A , has no all-negative 2×2 . Shattering three columns would require one row negative on all three and a distinct row negative on two of them, which creates such a submatrix. Therefore $\text{VC}(A) \leq 2$.

The homogeneous-rectangle theorem of Alon et al. [4], in the formulation checked in Hatami et al. [24, Theorem 1.9], states that an $a \times b$ sign matrix of sign-rank at most d contains a monochromatic rectangle with each side at least the corresponding side length divided by 2^{d+1} . Apply Lemma 2.2, since sign-rank is unchanged by repetition. It follows that

$$\text{rect}(F) \geq 2^{-2d-2}. \tag{B.1}$$

Fix a product distribution and such a rectangle of F . If its color is positive, it survives every monotone flip. If its color is negative, one side is a singleton, by the absence of a negative 2×2 . Partition the other side according to its sign in A ; one part retains at least half its mass. Thus

$$\text{signrank}(F) \leq d, \quad F \text{ has no negative } 2 \times 2 \implies \text{rect}(A) \geq 2^{-(2d+3)}. \tag{B.2}$$

For $d = 6$ this is 2^{-15} .

For the counting bound, we use the strict-pattern formulation of Warren's theorem [33] given in Alon et al. [5, Theorem 21 and Lemma 22]: $s \geq v$ real polynomials of degree at most k in v variables have at most $(4eks/v)^v$ strict sign patterns. A rank-at-most- d $K \times K$ matrix factors as a $K \times d$ matrix times a $d \times K$ matrix. Its K^2 entries are degree-two polynomials in $2Kd$ variables. Therefore, for $1 \leq d \leq K/2$,

$$\#\{B \in \{-1, +1\}^{K \times K} : \text{signrank}(B) \leq d\} \leq (4eK/d)^{2Kd} \leq 2^{2KdB_0}. \quad (\text{B.3})$$

The random family has exactly 2^I equiprobable matrices. Taking $d = s_0$ bounds the probability by $2^{2Ks_0B_0-I} \leq 2^{-I/2}$. The range $s_0 \leq K/2$ follows from $I \leq K^2$ and $B_0 > 1$; when $s_0 = 0$ the event is empty. Finally $I \geq N^4$ and $K = 2N^3$ give the asymptotic lower bound. Apply Theorem 3.1 for the learner. $\square$

C Proofs for cap mixtures and proper learning

C.1 Proof of Theorem 5.1

Proof of Theorem 5.1. We first make the once-only perturbation precise. Number the rows $j = 1, \dots, K$. Replace a row vector v_j by

$$v_j(t) = v_j + t(1, j, \dots, j^{d-1}).$$

Choose a positive rational upper bound t_0 so that every dot product retains its strict sign for $0 < t < t_0$. The finitely many original rational dot products and the perturbation dot products give such a bound by direct comparison. For each d -subset of rows, its determinant is a polynomial in t of degree d whose leading coefficient is a nonzero Vandermonde determinant. At most $d \binom{K}{d}$ values are forbidden. Testing that many plus one rational grid values in $(0, t_0)$ finds a valid value, with polynomial bit length and polynomial work for fixed d . When $K < d$, no perturbation is needed: use the evaluation-coordinate construction below for every survivor set. For $K \geq d$, the same perturbation works for every later survivor set.

Suppose there are $k > d$ surviving rows. Any proper j -dimensional subspace contains at most j of them, strictly less than kj/d . The radial-isotropy proof of Theorem 7.4 therefore applies; its convex formulation is also proved in Appendix J. Transform and normalize the rows to vectors $v_h \in S^{d-1}$ with

$$\mathbb{E}_\nu v_h v_h^\top = I_d/d.$$

Transform points contragrediently and normalize them to z_x . All template signs remain strict. If $k \leq d$, retain *all* rows and replace the factorization by evaluation coordinates: let $\tilde{z}_x = (\langle v_h, u_x \rangle)_{h \in V}$ and use row vectors $e_h \in \mathbb{R}^k$. Every coordinate of $\tilde{z}_x$ is nonzero by strictness, so its normalization is defined and preserves every sign. For the uniform surviving-row prior, $k^{-1} \sum_{h \in V} e_h e_h^\top = I_k/k$. This is a new factorization, not an invertible map of the original ambient space. No target is discarded.

Write the effective dimension as $j \leq d$, put $\alpha^2 = 1/(2j)$ and $\beta = \sqrt{1 - \alpha^2}$, and sample u uniformly on S^{j-1} . Define

$$C_u = \{x : \langle u, z_x \rangle \geq \beta\}, \quad T_u^\pm = \{h : \pm \langle u, v_h \rangle > \alpha\}.$$

For $Z = \langle u, v_h \rangle^2 \in [0, 1]$, isotropy gives $\mathbb{E}_\nu Z = 1/j$. Since

$$\mathbb{E} Z \leq \alpha^2 + \Pr[Z > \alpha^2],$$

we have $\nu(T_u^+) + \nu(T_u^-) \geq 1/(2j)$. Choose the larger side, of mass at least $1/(4j)$.

The geometry is strict. If $s = \langle u, v \rangle > \alpha$ and $t = \langle u, z \rangle \geq \beta$, then

$$\langle z, v \rangle \geq ts - \sqrt{1 - t^2} \sqrt{1 - s^2} > 0,$$

because $s, t > 0$ and $s^2 + t^2 > 1$. Replacing v by $-v$ proves the negative-side assertion. Thus $C_u \times T_u^+$ is positive in F , and $C_u \times T_u^-$ is negative in F .

A positive rectangle survives in A . A negative rectangle has a singleton side. If its row side is $\{h\}$, partition $C_u \cap U$ according to $h(x)$ in A and choose either color with probability $1/2$. If its point side is $\{x\}$, partition the row side according to $h(x)$ and choose either color with probability $1/2$. Empty outcomes are allowed. The singleton-point case is essential and is not replaced by a singleton-row assumption. Conditional on $x \in C_u$, its expected weighted coverage after this split is at least half the chosen side's mass, hence at least $1/(8j)$. Rotational invariance gives probability c_j of cap membership. Therefore the pointwise bound is $c_j/(8j) \geq c_d/(8d)$. To verify the last inequality, couple the first coordinate of uniform spheres by normalizing an increasing list of independent standard Gaussians. Its absolute value decreases as dimension grows, while $\sqrt{1 - 1/(2j)}$ increases. Symmetry supplies the factor $1/2$, including the case $j = 1$.

For finite-bit computation, Appendix J gives a deterministic rational transform with normalized covariance at least $3I_j/(4j)$. The preceding second-moment argument then gives total side mass at least $1/(4j)$, selected-side mass at least $1/(8j)$, and split coverage at least $1/(16j)$ whenever a point belongs to the cap.

Use the finite direction set

$$W_j = \left\{ \frac{g}{\|g\|} : 0 \neq g \in (16j)^{-1} \mathbb{Z}^j \cap [-1, 1]^j \right\}, \quad J_j = |W_j| \leq (32j + 1)^j.$$

Repetitions of a direction cause no difficulty. Rounding a unit vector coordinatewise produces g with $\|g - z\| \leq 1/(32\sqrt{j})$; normalization changes it by at most twice this amount. Thus $\langle z, g/\|g\| \rangle \geq 1 - 1/(512j) > \beta$, where the last inequality follows from $\beta \leq 1 - 1/(4j)$. Every point is in at least one cap. Uniformly sampling W_j and the split color yields coverage at least $1/(16jJ_j)$. The common rational lower bound

$$r_d = \frac{1}{16d(32d + 1)^d} \tag{C.1}$$

works in all effective dimensions. All membership comparisons are rational: for example $\langle z, g/\|g\| \rangle \geq \beta$ means that the unnormalized dot product is positive and

$$\langle z_{\text{raw}}, g \rangle^2 \geq (1 - 1/(2j)) \|z_{\text{raw}}\|^2 \|g\|^2.$$

The row-side tests use the analogous strict squared comparison. Consequently no exact sampling of a continuous direction is required. Theorem 3.1 with r_d proves the randomized finite-bit bound.

For determinism, enumerate the at most $2J_j$ outcomes and query each $D(S)$, using the same tolerance and stopping check as Theorem 3.1 with r_d . Choose the outcome maximizing $b\hat{s}$, where $b = \nu(T)$. Coverage implies that some outcome has $bs \geq r_d u$. The chosen one has

$$bs \geq r_d u - 2\tau, \quad ab \geq r_d - 2\tau/u \geq 3r_d/4.$$

Thus both a and b are at least $3r_d/4$. Query its mismatch mass and peel or eliminate exactly as before. The target survives, and all error bounds remain valid. Each stage has progress at least $3r_d/4$, whereas the deterministic total-progress bound is B . There cannot be more than $\lceil 4B/(3r_d) \rceil + 1$ nonstopping stages. There are $O_d(1)$ queries per stage. This proves the deterministic claim, without fixing a successful private seed after seeing D . $\square$

C.2 Proof of Theorem 5.2

Proof of Theorem 5.2. Set $\theta = \epsilon/4$ and run RECTIFY at accuracy θ , stopping at its first proposed negative peel. Until then all assigned labels, and its fallback, are positive. If it stops normally without

a negative peel, output the all-positive member of H . The error is at most $\theta/2$. Its budget-failure probability is at most $\theta/2$.

Suppose a negative rectangle $S \times T$ reaches a peeling answer. The acceptance calculation in Theorem 3.1 and $u > \theta/8$ give

$$D(S) > 5r\theta/128.$$

For $P_\theta = 1 + \lceil 4r^{-1} \ln(8/\theta) \rceil$ and $\tau = \theta/(24P_\theta)$,

$$\frac{D(S)}{\tau} > \frac{15}{16}rP_\theta > \frac{15}{4} \ln(8/\theta) > 18,$$

since $\theta < 1/16$. A peeling answer implies $D(S \cap \{h^* = +1\}) \leq 3\tau$. Hence the negative mass in S exceeds 15τ .

If $S = \{p\}$, then $h^*(p) = -1$ and p is known. Otherwise the negative rectangle has a single row h_0 . Query its full loss, accepting it when the answer is at most $\epsilon/2$. An accepted candidate has true loss at most $\epsilon/2 + \tau$. If it is rejected, $h_0 \neq h^*$, since the target's answer would be at most τ . The two rows share at most one negative point. Therefore the negative measure in S is concentrated at a unique point p and has mass greater than 15τ . Locate p by binary search over an arbitrary ordering of S : for a half $B \subseteq S$, query $D(B \cap \{h^* = -1\})$. The answer is at most τ when $p \notin B$, and greater than 14τ when $p \in B$. Threshold 4τ safely determines the half. This takes at most $\lceil \log_2 |X| \rceil$ queries.

Now let $V_p = \{h : h(p) = -1\}$. Its negative supports away from p are pairwise disjoint; otherwise two rows share two negative points. Query the target's negative mass away from p ,

$$\beta = D(\{x \neq p : h^*(x) = -1\}).$$

If the answer exceeds $\epsilon/4$, then $\beta > \epsilon/4 - \tau$. For any half $W \subseteq V_p$, define

$$Q_W(x, y) = \mathbf{1}_{\{y=-1\}} \mathbf{1}_{\{x \neq p\}} \mathbf{1}_{\{x \in \bigcup_{h \in W} \{h=-1\}\}}.$$

Its expectation is exactly β if $h^* \in W$ and exactly zero otherwise. Thus threshold $\epsilon/8$ distinguishes the two cases, since $\tau < \epsilon/16$. Binary search finds the target row in at most $\lceil \log_2 K \rceil$ queries. If two indexed rows coincide and contain only p as a negative point, this branch cannot occur; in the positive-tail branch the target row is unique.

If the answer for β is at most $\epsilon/4$, then $\beta \leq \epsilon/4 + \tau$. Test up to

$$J = \lceil 8/\epsilon \rceil + 1$$

distinct members of V_p , again accepting an estimated full loss at most $\epsilon/2$. If fewer than J members exist, these tests include the target. Otherwise their negative masses outside p sum to at most one, by disjointness. One tested row h has such mass at most $1/J < \epsilon/8$. It agrees at p , and therefore

$$L_{D, h^*}(h) \leq \beta + D(\{x \neq p : h(x) = -1\}) < 3\epsilon/8 + \tau.$$

Its answer is less than $\epsilon/2$, because $2\tau < \epsilon/8$. Some test consequently passes. Every accepted row has loss at most $\epsilon/2 + \tau$.

All nonfailure outcomes are proper and have loss at most $\epsilon/2 + \tau$. Any budget failure can output the all-positive row, with loss at most one. Expected loss is bounded by

$$\epsilon/2 + \tau + \theta/2 < \epsilon.$$

The extra query counts give (5.2). Using the deterministic net implementation removes the budget-failure event and leaves the same proper completion. Its computation consists of cap tests, row-label tests, and unions of explicitly listed row supports, all polynomial in the table size. $\square$

D Proof of the order-median learner

D.1 Proof of Lemma 5.3

Proof of Lemma 5.3. Start with the bracket $[L - 1, H]$, regarding the lower endpoint as a negative comparison and the upper as a positive comparison. The corresponding true masses are zero and β , so both satisfy the claimed endpoint inequalities. At an interior query, an answer at least $\hat{\beta}/2$ implies true mass at least $\beta/2 - 3\zeta/2$; an answer below it implies true mass below $\beta/2 + 3\zeta/2$. Preserve the respective endpoint inequalities during binary search. They remain valid even if different valid answers do not form a monotone sequence. At termination the endpoints are $v - 1, v$, proving the first two assertions.

In the light-atom case, $m \leq \hat{\beta}/8 + \zeta \leq \beta/8 + 9\zeta/8$. Add this to the upper bound on $\mu^-\{f < v\}$ to obtain (5.3). In the other case $m > \hat{\beta}/8 - \zeta \geq \beta/8 - 9\zeta/8 > \beta/9$, since $\beta \geq 512\zeta$. $\square$

D.2 Proof of Theorem 5.4

Proof of Theorem 5.4. Query β , obtaining $\hat{\beta}$. If $\hat{\beta} \leq \epsilon/4$, return the all-positive row $(a, b) = (1, 2N^2)$. Its loss is at most $\epsilon/4 + \zeta$. Otherwise

$$\beta > \epsilon/4 - \zeta > \epsilon/8, \quad \beta > 1023\zeta > 512\zeta.$$

All subsequent negative-measure queries are unconditioned indicators and have error at most ζ .

First take a noisy median v_x of x . If its atom is heavy in Lemma 5.3, it consists of one negative point: a nonvertical target line has at most one point with a given x coordinate. Binary queries $\mu^-\{x = v_x, y \leq t\}$ locate its y coordinate. Each such query has true mass either zero or more than $\beta/9$. Threshold $\hat{\beta}/32$ distinguishes these cases despite error ζ : it exceeds ζ and is less than $\beta/9 - \zeta$. We then use the heavy-point completion described below.

Otherwise fix the prefix $P = \{x \leq v_x\}$, whose mass satisfies (5.3). Maintain an integer interval containing the unknown slope $a^* \in [1, N]$. At its midpoint a , take a noisy median v of

$$f_a(x, y) = y - ax, \quad 1 - N^2 \leq f_a(x, y) \leq 2N^2.$$

On the target line this is $(a^* - a)x + b^*$. Query the mass m of its median atom.

Suppose that atom is heavy. Query its intersection mass l with P . If both $\hat{l} \geq \hat{\beta}/8$ and $\hat{m} - \hat{l} \geq \hat{\beta}/8$, then $a = a^*$ and $v = b^*$. Indeed, for $a \neq a^*$, f_a is injective on the target line, so its atom is a single point and lies wholly on one side of P . One of the true split masses is zero. Its directly observed value is at most ζ , or its difference of two observed values at most 2ζ , both less than $\hat{\beta}/8$. Conversely, when $a = a^*$, the atom has mass β and the median value is exactly b^* . The two true split masses are at least $\beta/2 - 3\zeta/2$ and $3\beta/8 - 21\zeta/8$ by (5.3). Their observed lower bounds exceed $\hat{\beta}/8$, using $\beta > 512\zeta$. Thus the test accepts precisely the constant-residual case. Return the exact target row in that case. If the split test fails, the heavy atom is a single negative point; locate its x coordinate by binary queries restricted to $\{f_a = v\}$, then put $y = ax + v$ and use the heavy-point completion.

It remains to handle a light residual atom. The set $Q = \{f_a \leq v\}$ also satisfies (5.3). If $a^* > a$, the restrictions of P, Q to the target line are prefixes of the same order. Consequently

$$\mu^-(P \cap Q) \geq \beta/2 - 3\zeta/2.$$

If $a^* < a$, they are a prefix and a suffix. Their union covers the entire negative support whenever they overlap. Thus, whether or not they overlap,

$$\mu^-(P \cap Q) \leq \max\{0, \mu^-(P) + \mu^-(Q) - \beta\} \leq \beta/4 + 21\zeta/4.$$

Query this intersection and compare its answer to $3\hat{\beta}/8$. In the first case the answer is at least $\beta/2 - 5\zeta/2 > 3\hat{\beta}/8$; in the second it is at most $\beta/4 + 25\zeta/4 < 3\hat{\beta}/8$. Both inequalities follow from

$\beta > 512\zeta$. Equality of slopes is impossible in the light case, because the entire negative mass would be one residual atom. Retain the appropriate strict half of the slope interval. After $O(\log N)$ stages only a^* remains. A median query search for its constant residual gives b^* , so return h_{a^*, b^*} . Each slope stage uses $O(\log N)$ queries. This is the all-light algorithm.

Heavy-point completion. We have located an actual negative point p . Query the negative tail

$$\beta_p = \mu^-(X \setminus \{p\}).$$

If its answer exceeds $\epsilon/4$, then $\beta_p > \epsilon/4 - \zeta$. For a half-interval of slopes in the star at p , query negative mass away from p on the union of those lines. Every negative point other than p determines the same unique target slope, so this query is exactly zero or β_p . Threshold $\epsilon/8$ distinguishes the cases. Binary search identifies the target in $O(\log N)$ queries. The union is succinctly evaluable: for a query point $q \neq p$, test $q_x \neq p_x$, divisibility of $q_y - p_y$ by $q_x - p_x$, and membership of the quotient in the chosen slope interval. This requires no enumeration and no evaluation of flips.

If the tail answer is at most $\epsilon/4$, the improper learner returns the point predictor that is negative only at p . Its loss is $\beta_p \leq \epsilon/4 + \zeta$. This proves the deterministic improper statement for every evaluable flip matrix.

For proper output, put

$$C_0 = \max\{64\lceil\log_2(2N^3)\rceil, \lceil 128/\epsilon \rceil\}.$$

If $t(p) \leq C_0$, enumerate its slopes, discard rows positive at p by an entry evaluation, and query the full loss of each remaining row until an answer is at most $\epsilon/2$. The target is among them and must pass. A passing row has actual loss at most $\epsilon/2 + \zeta$.

If $t(p) > C_0$, star-regularity supplies at least $t(p)/4$ rows negative at p . The sets of negative points of different star rows outside p are disjoint. At most $8/\epsilon$ rows can therefore have D -mass greater than $\epsilon/8$ on their outside negative sets. At least $t(p)/4 - 8/\epsilon \geq t(p)/8$ rows are negative at p and have outside negative mass at most $\epsilon/8$. Every such row has loss at most

$$\beta_p + \epsilon/8 \leq 3\epsilon/8 + \zeta,$$

and its loss answer is at most $3\epsilon/8 + 2\zeta < \epsilon/2$, regardless of the oracle. Sample

$$J = \lceil 16 \ln(4/\epsilon) \rceil$$

independent star slopes as follows: draw $q = \lceil \log_2 t(p) \rceil$ unbiased bits, interpret them as an integer, reduce modulo $t(p)$, and add one. Every slope has probability at least $1/(2t(p))$, so a guaranteed-passing row has probability at least $1/16$ in each trial. Evaluate its label at p and test negative rows as above. The chance that no guaranteed-passing row is sampled is at most $(1 - 1/16)^J \leq \epsilon/4$. On that event return the all-positive row. All other outputs have error at most $\epsilon/2 + \zeta$. The expected loss is at most $\epsilon/2 + \zeta + \epsilon/4 < \epsilon$.

There are $O(n^2)$ median queries, $O(n)$ locating and star-search queries, and at most $O(n + 1/\epsilon + \log(1/\epsilon))$ candidate tests. Every predicate uses polynomially many n -bit integer operations and at most one entry evaluation. The output is either a row index with its evaluation algorithm or, in the improper case, a point predictor. The finite answer rounding specified before Lemma 5.3 completes the bit-complexity assertion. $\square$

E Proofs for the explicit families

E.1 Proof of Lemma 6.1

Proof of Lemma 6.1. When $D_k \geq 1$, we have $k \geq 32$ and $1 \leq D_k \leq k \leq K_k/2$, so the range condition in (B.3) holds. We use $k \geq 32$ only to bound the logarithm, not as the first positive-floor threshold.

Thus

$$\log_2(4eK_k/D_k) \leq \log_2(8ek^3) \leq 4\log_2 k.$$

The strict-pattern bound (B.3) therefore gives at most

$$2^{2K_k D_k \cdot 4\log_2 k} \leq 2^{(2/3)k^4}$$

low-rank tables. For $k \geq 3$, $(k+1)^2/(4k^2) \leq 4/9 < 1/2$, so $I_k \geq (3/2)k^4$. Dividing the preceding count by 2^{I_k} gives at most $2^{-5k^4/6} \leq 2^{-k^4/2}$. If $D_k = 0$, no nonempty strict sign matrix has that rank.

For the decision algorithm, query the incidence bits and fill the known off-incidence entries. The condition is equivalent to feasibility of

$$A_{ij} \sum_{s=1}^{D_k} U_{is} V_{sj} \geq 1 \quad (i, j \in [K_k]).$$

Indeed a strict factorization can be scaled to have all signed scores at least one, since there are finitely many positive signed scores. The converse is immediate. There are $2K_k D_k$ real variables and K_k^2 degree-two polynomial inequalities with bounded integer coefficients. We use the singly exponential existential-real consistency bound in the formulation of Vorobjov [32], attributed there to Renegar [31]: $(sd)^{O(v)} L^{O(1)}$ bit operations for s polynomials of degree at most d , v variables, and coefficient bit length L . Substitution gives $2^{O(k^3 D_k \log k)}$. This is a use of the classical decision theorem, not a practical algorithm for the large full table. $\square$

E.2 Proof of Theorem 6.2

Proof of Theorem 6.2. The evaluation claim follows from integer arithmetic and one PRF evaluation. The rectangle and proper-learning bounds hold for every monotone flip, without any randomness assumption on its entries.

Take the subgrid scale $k = \lceil cn^c \rceil$, which is at most N for sufficiently large n . Restrict both the full row set and point set to $[k] \times [2k^2]$. The template and flip law restricted there are exactly the scale- k construction. An oracle distinguisher reads its incidence bits and accepts precisely when its sign-rank is at most D_k . Its time is

$$2^{O_c(n^{4c})} \leq 2^{n^{4c+2}} \leq 2^{\sigma^\delta}$$

for sufficiently large n . Under a uniform function its acceptance probability is at most $2^{-k^4/2}$ by Lemma 6.1. Under the PRF it is therefore at most $2^{-k^4/2 + \text{negl}(\sigma)}$, which is negligible in n . Restricting a sign realization cannot increase its rank. Consequently, except on that event, the full dimension is at least $D_k + 1 > k/(24\log_2 k)$.

For large n , $k \geq cn^c$ and $\log_2 k \leq c\log_2 n + \log_2(2c) \leq (5/4)c\log_2 n$. Hence

$$D_k + 1 > \frac{k}{24\log_2 k} \geq \frac{n^c}{30\log_2 n}.$$

This proves the displayed constant. The choice $k = n^c$ alone would instead give $n^c/(24c\log_2 n)$; its factor c cannot be discarded. Also $N = 2^{(n-1)/3}$, rather than $2^{n/3}$, respects the fixed definition $n = \log_2 |X|$. $\square$

E.3 Proof of Lemma 6.4

Proof of Lemma 6.4. At a fixed point with t incident rows, the number of negatives is a sum of t independent fair bits. For $\lambda = \ln 3$,

$$\Pr[Z < t/4] \leq e^{\lambda t/4} \left(\frac{1 + e^{-\lambda}}{2} \right)^t = \left(\frac{2}{3^{3/4}} \right)^t \leq e^{-t/16}.$$

For the last elementary inequality, $\ln(3^{3/4}/2) > 1/16$; for example $3^3/2^4 = 27/16 > e^{1/4}$, since $e^{1/4} \leq (1 - 1/4)^{-1} = 4/3 < 27/16$. If $t \geq 64\lceil n \rceil$, the failure probability is at most e^{-4n} . There are at most 2^n points, so a union bound gives $e^{-(4-\ln 2)n} \leq e^{-3n}$.

A distinguisher can read every incidence bit and test all stars in time $2^{O(n)}$. This is below 2^{σ^δ} for the stated key length. The random-function bound plus PRF indistinguishability proves the claim. Polynomial growth of σ in n makes e^{-3n} negligible in the key length as well. $\square$

E.4 Proof of Corollary 6.5

Proof of Corollary 6.5. Intersect the high-rank key event from Theorem 6.2 with the star-regular event from Lemma 6.4. Their two negligible exceptional probabilities have negligible sum. Apply Theorem 5.4 and the polynomial-time entry evaluator. $\square$

E.5 Proof of Proposition 6.3

Proof of Proposition 6.3. Take $k = \lceil cn^c \rceil$ and store the I_k independent incidence bits of the scale- k subgrid. Set every other template incidence positive. The proof of Theorem 6.2, with independent bits in place of the PRF, gives the lower bound and exceptional probability from Lemma 6.1. Entry evaluation checks subgrid membership and, only there, reads the relevant advice bit.

Collapse all points outside the subgrid into one positive point. The effective domain has $2k^3 + 1$ points and the effective class at most $2k^3 + 1$ rows; every row outside the subgrid is all-positive. A strict template representation in dimension seven is obtained from the original six coordinates by adjoining a coordinate that is zero on old points, one on the added point, and one on all row vectors. The added point vector has its other coordinates zero. An all-positive row already occurs in the subgrid, so the collapsed rows require no additional row sign pattern. Adding it preserves the no-negative- 2×2 property.

Apply the deterministic cap and proper constructions to this polynomial-size effective table. Every effective SQ is simulated by composing its query with the known collapsing map, so its expectation under an arbitrary original marginal is exactly the expectation under the pushforward marginal. The proper output extends by +1 outside the subgrid and belongs to the original class. The query count is $O_\epsilon(\log k) = O_{c,\epsilon}(\log n)$, and the table computation is polynomial in n for fixed c . $\square$

F Spectral, communication and representation proofs

F.1 Proof of Theorem 7.1

Proof of Theorem 7.1. Fix a product distribution and a protocol of expected cost at most $c > 0$. Markov's inequality gives probability at least $1/2$ to leaves of depth at most $2c$. There are at most $2^{\lfloor 2c \rfloor}$ such leaves: their transcript words are prefix-free, and each can be extended to a distinct word of length $\lfloor 2c \rfloor$. One of their rectangles has mass at least $2^{-\lfloor 2c \rfloor - 1} \geq 2^{-2c-1}$. A zero-cost protocol has a monochromatic support rectangle of mass one, so the same claimed weaker bound holds. Taking the infimum over product distributions proves the first inequality. The finite domain permits a minimum expected-cost exact protocol: redundant repeated-state branches can be removed, bounding depth by a finite function of the domain sizes. Alternatively, use costs within arbitrarily small additive slack and pass to the limit.

For the converse put $r = \text{rect}(H)$. At any current rectangular cell, condition the original product distribution on that cell. Its positive-mass conditional law is still product, and the restricted matrix still has rectangle ratio at least r . Fix a monochromatic rectangle of conditional mass at least r . The parties exchange their two membership bits. If both are one, output its color. Otherwise continue

in the indicated one of the three remaining rectangular cells. Every nonempty continuing cell is a proper subcell: a positive-mass rectangle has both sides nonempty, and either a row or point has been excluded. Thus this exact protocol terminates after finitely many stages on every positive-support pair. Zero-mass cells can be completed by any finite exact protocol, without expected cost.

Conditional on reaching a stage with positive probability, its termination probability is at least r . By induction its survival probability after t stages is at most $(1 - r)^t$. Summing the tail probabilities, the expected number of stages is at most $1/r$, and each costs at most two bits. This proves the second inequality. This converse is the protocol of Hatami et al. [24, Proposition 3.14], included here with the conditioning and termination details. Substitute $\rho = 2^{2c+1}$ into Theorem 3.1 for the SQ bounds. $\square$

F.2 Proof of Lemma 7.2

Proof of Lemma 7.2. Let $T : \mathbb{R}^K \rightarrow L^2(P_0)$ send b to $\sum_j b_j r_j$. Then $T^*T = \Gamma$. For every unit vector b ,

$$|\langle b, T^*a \rangle| = |\langle Tb, a \rangle| \leq \|Tb\|_2 \|a\|_2 \leq \sqrt{\Lambda} \|a\|_2.$$

Taking the supremum over b and squaring proves (7.1). Each selected summand in (7.2) is greater than τ^2 ; discarding nonnegative unselected summands proves the result. The same argument, with weak inequalities, proves the final statement even when $\tau = 0$. $\square$

F.3 Proof of Theorem 7.3

Proof of Theorem 7.3. First $\Lambda > 0$. Indeed, with $a_j(x, y) = y h_j(x)$ we have $\|a_j\|_{L^2(P_0)} = 1$ and $\langle r_j, a_j \rangle = \mathbb{E}_{P_j} a_j - \mathbb{E}_{P_0} a_j = 1$. Thus $\|r_j\|_2 \geq 1$ and $\Lambda \geq \Gamma_{jj} \geq 1$.

For a fixed private seed, run the learner on the reference responses $\mathbb{E}_{P_0} q$. Write the resulting queries as $q_1, \dots, q_m$ and the leaf output as g_0 . This is a counterfactual tree path; the reference need not be realizable. For target j , choose the following memoryless oracle, which does not know the private seed:

$$O_j(q) = \begin{cases} \mathbb{E}_{P_0} q, & |\mathbb{E}_{P_j} q - \mathbb{E}_{P_0} q| \leq \tau, \\ \mathbb{E}_{P_j} q, & \text{otherwise.} \end{cases}$$

It is valid for P_j on every query. If its run first departs from the reference transcript, that departure occurs at one of the reference queries with $|\langle r_j, q_t \rangle| > \tau$. Lemma 7.2 and a union bound give, for this fixed seed,

$$\frac{1}{K} |\{j : \text{the transcript departs from the reference}\}| \leq \frac{m\Lambda}{K\tau^2}.$$

For the reference leaf, define $c_j = \mathbb{E}_{P_j}[y g_0(x)]$. Since $\mathbb{E}_{P_0}[y g_0(x)] = 0$, Lemma 7.2 gives

$$\frac{1}{K} \sum_j |c_j| \leq \sqrt{\frac{1}{K} \sum_j c_j^2} \leq \sqrt{\Lambda/K}.$$

Let s_j be the correlation of the actual output under O_j . On a nondeparting transcript it equals c_j ; on a departing transcript it is at most 1. Therefore $s_j \leq |c_j| + \mathbf{1}_{\text{departure for } j}$. Average this inequality over j and then over the private seed. The learner's guarantee gives an average correlation at least $1 - 2\epsilon$. This proves (7.3). Notice that we did not choose a different oracle after seeing the seed.

Finally write $M = m/\tau^2$. From (7.3), $R/2 \leq \sqrt{R} + M$. The square $(\sqrt{R} - 2)^2 \geq 0$ gives $\sqrt{R} \leq R/4 + 1$. Combining yields (7.4). $\square$

F.4 Proof of Theorem 7.4

Proof of Theorem 7.4. Let $d = \text{signrank}(M)$, with $M = \text{sign } A$ in the first assertion. Choose a strict representation $M_{xj} = \text{sign}\langle z_x, v_j \rangle$ in $\mathbb{R}^d$. If $d \geq K$, both conclusions are immediate: $\|A\|_{\text{op}} \geq \sqrt{N}$ from any column, while $G_{jj} = 1$ implies $\|G\|_{\text{op}} \geq 1$. We henceforth have $1 \leq d < K$.

Perturb the finitely many v_j within sufficiently small open neighborhoods so that signs are unchanged and every d of them are independent. Such a perturbation exists: for each d -subset the determinant is a nonzero polynomial in the vector coordinates, and a finite union of its zero sets has empty interior. To justify this elementary fact, a nonzero univariate polynomial has finitely many roots; induction on the number of variables, by viewing the polynomial as a polynomial in the last variable, shows that a multivariate polynomial cannot vanish on a nonempty open box. Apply this to the product of the finitely many determinant polynomials.

We prove that there is an invertible T such that the normalized vectors $u_j = Tv_j/\|Tv_j\|$ satisfy

$$\sum_{j=1}^K u_j u_j^\top = (K/d)I_d. \quad (\text{F.1})$$

For every $1 \leq r \leq d$, there is a common $\delta > 0$ such that at most $d - r$ of the v_j have projection norm less than δ onto any r -dimensional subspace. In fact, any $d - r + 1$ of the vectors are independent, so cannot all belong to the orthogonal complement of an r -dimensional subspace. Their maximum projection norm is therefore positive for every such subspace. Its minimum is positive by compactness of the space of orthogonal projection matrices of rank r ; this space is closed and bounded. Take the minimum over finitely many subsets and r .

For positive definite Q with $\det Q = 1$, minimize

$$F(Q) = \sum_{j=1}^K \log(v_j^\top Q v_j).$$

Write the decreasing eigenvalues of Q as $e^{s_1} \geq \dots \geq e^{s_d}$, where $\sum_i s_i = 0$. For each r , at least $K - d + r$ of the quadratic forms are at least $\delta^2 e^{s_r}$. Order the $K \log$ quadratic forms increasingly. For the first d entries use respectively the bounds $\log \delta^2 + s_d, \dots, \log \delta^2 + s_1$; each of the remaining $K - d$ entries is at least $\log \delta^2 + s_1$. Thus

$$F(Q) \geq K \log \delta^2 + (K - d)s_1.$$

Since $K > d$, every sublevel set has bounded largest eigenvalue. The determinant condition then bounds the smallest eigenvalue away from zero. Sublevel sets are compact, so a minimizer Q exists. For a symmetric traceless matrix B , the path $Q(t) = Q^{1/2} \exp(tB) Q^{1/2}$ retains determinant one. Its derivative at zero gives

$$0 = \sum_j \frac{v_j^\top Q^{1/2} B Q^{1/2} v_j}{v_j^\top Q v_j} = \text{tr} \left(B \sum_j u_j u_j^\top \right), \quad u_j = \frac{Q^{1/2} v_j}{\|Q^{1/2} v_j\|}.$$

A symmetric matrix orthogonal to every traceless symmetric matrix is scalar: subtract its scalar trace part and take that difference as B . Taking traces gives (F.1), with $T = Q^{1/2}$.

Replace each row vector by the normalization of $T^{-\top} z_x$, denoted e_x , so $\|e_x\| = 1$ and all signs are preserved. Let U be the matrix with columns u_j . For every row of A ,

$$\langle e_x, U A_{x,\cdot}^\top \rangle = \sum_j |A_{xj}| |\langle e_x, u_j \rangle| \geq \sum_j \langle e_x, u_j \rangle^2 = K/d.$$

Here $|\langle e_x, u_j \rangle| \leq 1$ was used. Consequently,

$$NK^2/d^2 \leq \|UA^\top\|_F^2 \leq \|A\|_{\text{op}}^2 \|U\|_F^2 = K\|A\|_{\text{op}}^2.$$

This proves (7.6). For M , use the same row inequality, square, and average under μ :

$$K^2/d^2 \leq \sum_x \mu(x) \|UM_{x,\cdot}^\top\|^2 = \text{tr}(UGU^\top) \leq \|G\|_{\text{op}} \text{tr}(UU^\top) = K\|G\|_{\text{op}}.$$

The last inequality follows by diagonalizing the positive semidefinite G , whose eigenvalues lie between zero and its operator norm. This proves (7.7). $\square$

F.5 Proof of Theorem 7.5

Proof of Theorem 7.5. Points are one-dimensional subspaces of $\mathbb{F}_q^3$; lines are kernels of nonzero linear functionals, modulo nonzero scalar multiples. There are $(q^3 - 1)/(q - 1) = N$ of each. A line contains $(q^2 - 1)/(q - 1) = k$ points. Two distinct points span a unique two-dimensional subspace, hence lie on one common line. Two distinct kernels intersect in one one-dimensional subspace. These facts also show each point lies on k lines.

Let P be the $N \times N$ incidence matrix. Then

$$P\mathbf{1} = k\mathbf{1}, \quad P^\top P = qI + \mathbf{1}\mathbf{1}^\top.$$

Define $A = (N/k)P - \mathbf{1}\mathbf{1}^\top$. Its positive entries equal $N/k - 1 = q^2/(q + 1) \geq 1$, and its negative entries equal -1 . Direct multiplication gives

$$A\mathbf{1} = 0, \quad A^\top A = (N/k)^2 q \left(I - \frac{\mathbf{1}\mathbf{1}^\top}{N} \right).$$

Indeed, the coefficient of $\mathbf{1}\mathbf{1}^\top$ before simplification is $N^2/k^2 - N = -Nq/k^2$, since $k^2 = N + q$. Thus $\|A\|_{\text{op}} = (N/k)\sqrt{q}$, and Theorem 7.4 gives the lower bound in (7.8).

For the upper bound, enumerate the points by distinct integers $1, \dots, N$ and use the common embedding $\phi(t) = (1, t, \dots, t^{2k})$. For a positive set S of size k , the polynomial

$$p_S(t) = \frac{1}{2} - \prod_{a \in S} (t - a)^2$$

is positive on S and negative on all other domain points: away from S the product is a positive integer. Its coefficient vector is the required weight vector. Hence $\text{dc} \leq 2k + 1$.

Now prove (7.9). Write $K = |J|$, and restrict P to the selected columns. If $K < 16$, the Gram matrix has diagonal entries one, so its norm is at least one and the result follows. Suppose $K \geq 16$ and let r_x be the number of selected lines containing point x . If $r_x < K/4$ for every x , take the unit vector $a = \mathbf{1}/\sqrt{K}$. At each point,

$$|\sum_{j \in J} a_j h_j(x)| = |2r_x - K|/\sqrt{K} > \sqrt{K}/2.$$

Its Gram Rayleigh quotient is at least $K/4$. Otherwise choose x_0 incident with $r \geq K/4$ selected lines and let S be those r lines. At x_0 all these labels are positive; at any other point at most one is positive, since two points have a unique common line. For the unit vector $a = \mathbf{1}_S/\sqrt{r}$,

$$|\sum_j a_j h_j(x)| \geq (r - 2)/\sqrt{r} \geq \sqrt{r}/2 \quad \text{for all } x,$$

since $r \geq 4$. Its Rayleigh quotient is at least $r/4 \geq K/16$, for every μ . This proves (7.9), including marginals concentrated on a single point.

The claimed learner queries $q_j(x, y) = y h_j(x)$ for every line and outputs a hypothesis with maximum answer. The true target has correlation one, hence receives answer at least $1 - \tau$. A maximizing hypothesis has true correlation at least $1 - 2\tau$, and loss at most τ , regardless of the marginal and the oracle. Thus it is one distribution-independent learner.

We prove the fixed- D claim before excluding a common efficient learner. Let $\delta = \epsilon/2$ and sample a length- s sequence from the known D . For any pair of hypotheses at D -distance greater than δ , the probability of identical labels on the sequence is at most $(1 - \delta)^s \leq e^{-\delta s}$. A union bound over fewer than N^2 pairs is at most $N^2 e^{-\delta s} \leq e^{-1} < 1$. Consequently some sequence has the property that hypotheses with identical traces are within D -distance δ . Hardwire one such sequence and one representative hypothesis per trace into the learner. On at most s distinct sampled points there are at most $1 + s + \binom{s}{2}$ traces: the empty trace, at most s singleton traces, and at most one line through each pair of distinct sampled points. Every trace of size at least two is determined by one such pair. The representatives therefore form a D -cover of the claimed size. Query their label correlations and select the largest answer. A representative within δ of the true target has correlation at least $1 - 2\delta$. With tolerance $\tau = \epsilon/2$, the selected output has correlation at least $1 - 2\delta - 2\tau$ and error at most $\delta + \tau = \epsilon$. The hardwired cover depends on D ; this is not one learner serving all marginals.

To rule out a much cheaper common learner, choose for target line j the distribution

$$D_j(x) = \begin{cases} (2k)^{-1}, & P_{xj} = 1, \\ (2(N - k))^{-1}, & P_{xj} = 0. \end{cases}$$

Take the reference P_0 to be uniform on $X_q \times \{-1, +1\}$. With $c = k/(N - k)$, its residual likelihood ratios satisfy

$$r_j(x, +1) = A_{xj}, \quad r_j(x, -1) = -c A_{xj}.$$

For example the first formula is $2ND_j(x) - 1 = N/k - 1$ on the line and -1 elsewhere; the second is -1 on the line and $N/(N - k) - 1 = c$ elsewhere. Consequently

$$\begin{aligned} \Gamma &= \frac{1 + c^2}{2N} A^\top A, \\ \Lambda &= \frac{qN}{2} \left(\frac{1}{(q + 1)^2} + \frac{1}{q^4} \right), \\ R &= \frac{2}{q((q + 1)^{-2} + q^{-4})} \geq \frac{(q + 1)^2}{q} \geq q. \end{aligned}$$

The penultimate inequality uses $q^4 \geq (q + 1)^2$ for $q \geq 2$, which follows from $q^2 \geq q + 1$. By Theorem 7.3, $R \leq 4 + 4M$ for $M = m/\tau^2$. Therefore

$$\text{dc}(\mathcal{H}_q) \leq 2q + 3 \leq 2R + 3 \leq 11 + 8M \leq 19M,$$

where the final inequality uses $M \geq 1$. □

F.6 Proof of Theorem 7.6

Proof of Theorem 7.6. The grid has spacing $2/(B - 1) \leq \tau$, so nearest-grid rounding errs by at most $\tau/2$. The marginal tree has at most B^u leaves: suppress unary nodes and use induction on the number of remaining branching nodes per path. It has at most $(m + 1)B^u$ nodes, because every node has a descendant leaf and a depth between zero and m , and assigning each node one such pair is injective when the depth is fixed along a leaf's path. Collect the function b at every internal node and the Boolean output at every leaf. These are the common features $f_1, \dots, f_s$, where $s \leq (m + 1)B^u$ and each feature has absolute value at most one.

Fix D, h . Follow the tree by giving the exact constant response or rounding $\mathbb{E}_D a$ to the grid. If at some node $|\mathbb{E}_D h b| > \tau/2$, one of the signed features $b, -b$ has correlation greater than $\tau/2$. Otherwise every answer is a valid SQ response: the difference from the true expectation $\mathbb{E}_D a + \mathbb{E}_D h b$ is at most $\tau/2 + \tau/2$. A valid policy realizing this finite transcript extends to all other histories by exact answers. The deterministic guarantee then gives leaf correlation at least $1 - 2\epsilon > 1/2 \geq \tau/2$. We have proved

$$\forall D, \quad \max_{i, s_i \in \{-1, +1\}} \mathbb{E}_D[h(x) s_i f_i(x)] \geq \tau/2. \quad (\text{F.2})$$

Apply Lemma 2.1 with instance set X and actions the signed features (i, s_i) . Equation (F.2) gives a convex combination whose coordinate at every x is at least $\tau/2$. Its coefficients therefore define a common-feature linear representation with strict target product at every point. This proves (7.11). For correlational-only queries $a = 0$, hence $u = 0$.

For a randomized learner, fix a seed and instead retain all grid answers to each full query. There are at most B^m leaves. Use all leaf outputs as coordinates of an embedding, padding when needed. This distribution over embeddings depends only on the learner's seed, not on D, h . For fixed D, h , the policy rounding each true expectation to the grid is valid. Its output is one of the leaf features. Selecting that coordinate as w gives

$$\inf_w L_{D,h}(\text{sign}\langle w, \phi_\omega \rangle) \leq L_{D,h}(A_\omega^{O_{D,h}}).$$

Averaging over the seed proves (7.12). No seed is selected after observing D or h . $\square$

F.7 Proof of Proposition 7.7

Proof of Proposition 7.7. Let P have lines as rows and points as columns, with value one on incidences. Each row has p ones. Two rows with different slopes intersect once; rows with the same slope and different intercepts are disjoint. Hence

$$PP^\top = pI + J - \text{diag}(J_p, \dots, J_p).$$

Set $A = pP - J$. Direct multiplication using $PJ = pJ$ gives

$$AA^\top = p^2(pI - \text{diag}(J_p, \dots, J_p)), \quad \|A\|_{\text{op}}^2 = p^3.$$

For target line h , use probability $1/(2p)$ on each point of the line and $1/(2p(p-1))$ on each other point; labels are positive exactly on the line. Against the uniform fair-label reference, the likelihood-ratio residuals are $r_h(x, +1) = A_{hx}$ and $r_h(x, -1) = -A_{hx}/(p-1)$. Therefore

$$\Lambda = \frac{p}{2}(1 + (p-1)^{-2}), \quad R = p^2/\Lambda.$$

This proves the ratio and the expected-error consequence.

For the success-probability statement use the fixed, valid reference-following oracle from the proof of Theorem 7.3. For each fixed learner seed, at most fraction $m/(R\tau^2)$ of targets depart from its reference path. Let c_h be the correlation of the reference leaf with target h . The energy lemma gives $\sum_h c_h^2 \leq \Lambda$. Consequently at most fraction $1/(4\gamma^2 R)$ of targets have $c_h \geq 2\gamma$. The fraction of successful actual outputs is no larger than the sum of these two fractions. Average over the learner seed and use the assumed success probability on each target. This gives $\alpha \leq m/(R\tau^2) + 1/(4\gamma^2 R)$ and the stated inequality. Substitution of t yields $(2/3)p/t^2 - 1/4 = \Omega(\sqrt{p})$ in the cited weak regime. This comparison keeps both tolerance and success criteria explicit. $\square$

F.8 Proof of Theorem 7.8

Proof of Theorem 7.8. We prove the lower bound for any bipartite incidence graph whose degrees on both sides are at most $\Delta \geq 1$. Fix arbitrary product weights $\mu \times \nu$ and let p be their total incidence mass. If $p \geq 1/2$, some point has incident row mass at least $1/2$. At most Δ rows meet it, so one row has prior mass at least $1/(2\Delta)$. One of that row's two color classes has point mass at least $1/2$. This gives a monochromatic rectangle of mass at least $1/(4\Delta)$.

If $p < 1/2$, select each point independently with probability $t = 1/(2\Delta)$, obtaining S , and let T be all rows with no neighbor in S . The rectangle $S \times T$ is nonincident. For a nonincident pair (x, h) ,

$$\Pr[x \in S, h \in T] = t(1-t)^{\deg h} \geq t(1-\Delta t) = 1/(4\Delta).$$

The inequality $(1-t)^j \geq 1-jt$ follows by induction. Averaging the rectangle mass gives at least $(1-p)/(4\Delta) > 1/(8\Delta)$. Hence some rectangle has that mass. Taking $\Delta = q+1$ proves the universal lower bound.

For the upper bound choose uniform weights. Let $K = q^2 + q + 1$, $k = q + 1$, and let P be the incidence matrix. The identities $P^\top P = qI + \mathbf{1}\mathbf{1}^\top$ and $P\mathbf{1} = k\mathbf{1}$ give

$$C = P - (k/K)\mathbf{1}\mathbf{1}^\top, \quad \|C\|_{\text{op}} = \sqrt{q}.$$

A nonincident rectangle of sizes a, b satisfies $(k/K)ab \leq \sqrt{q}\sqrt{ab}$, so $ab/K^2 \leq q/k^2$. An incident rectangle has a singleton side and area at most k . Its mass is at most $1/K \leq q/k^2$, since $qK - k^2 = q^3 - q - 1 > 0$. This establishes the claimed upper bound on the infimum. $\square$

F.9 Proof of Theorem 7.9

Proof of Theorem 7.9. All integer inner products are odd and nonzero, so the identity embedding proves the dimension claim. Distinct parameter vectors give distinct rows: conditioning on all but coordinate i shows

$$\mathbb{E}_x[x_i \text{sign}\langle w, x \rangle] = w_i \Pr[Z_1 + \dots + Z_{n-1} = 0],$$

where the Z_j are independent fair signs and the probability is positive. The conditional difference of the two signs is two exactly when that even sum is zero, and is zero otherwise. Thus uniform parameters are also a uniform prior on the distinct hypotheses. We first bound every rectangle under this uniform product measure. A negative rectangle gives two subsets of the cube with all cross Hamming distances at least $t = (n+1)/2$. For a positive rectangle, negate one of its sides to get the same separation condition. Negation preserves cardinality.

Let A, B be nonempty subsets separated in this way. For uniform Z on the cube, set $f(Z) = \text{dist}_{\text{Ham}}(Z, A)$ and $s = \mathbb{E}f$. Changing one coordinate changes f by at most one. The elementary bounded-differences calculation proved in Appendix K gives

$$\Pr[f - s \leq -a] \leq e^{-2a^2/n}, \quad \Pr[f - s \geq a] \leq e^{-2a^2/n} \quad (a \geq 0).$$

Writing $\alpha = |A|/2^n$, $\beta = |B|/2^n$, if $0 \leq s \leq t$ then

$$\alpha\beta \leq \exp\{-2(s^2 + (t-s)^2)/n\} \leq e^{-t^2/n} \leq e^{-n/4}.$$

If $s \geq t$, already $\alpha \leq e^{-2t^2/n}$ and $\beta \leq 1$. Empty rectangles have zero mass. Taking the maximum under this one product measure proves the rectangle upper bound.

For learning, set $z = x/\sqrt{n}$, $\gamma = 1/n$, and optimize

$$F(v) = \mathbb{E}_D(\gamma - y\langle v, z \rangle)_+ \quad \text{over } \|v\| \leq 1.$$

The target $v^* = w^*/\sqrt{n}$ has $F(v^*) = 0$. At a current v , a valid subgradient is

$$g(v) = \mathbb{E}[-yz \mathbf{1}_{\{y\langle v, z \rangle < \gamma\}}], \quad \|g(v)\| \leq 1.$$

Query each coordinate using the bounded function $q_i(x, y) = -yx_i \mathbf{1}_{\{y\langle v, x/\sqrt{n} \rangle < \gamma\}}$ and divide its answer by $\sqrt{n}$. The resulting vector is $g(v) + e$ with $\|e\| \leq \tau$, regardless of the coordinatewise answer choices.

Start $v_0 = 0$, take $T = \lceil 100/(\epsilon^2 \gamma^2) \rceil$, step $a = T^{-1/2}$, and update by projection of $v_t - a(g_t + e_t)$ onto the unit ball. For any point v^* in that ball, projection cannot increase distance to v^* . Expanding the squared distance and using convexity yields

$$\frac{1}{T} \sum_{t \leq T} F(v_t) \leq \frac{1}{2aT} + \frac{a}{2}(1 + \tau)^2 + 2\tau \leq \frac{5}{2\sqrt{T}} + 2\tau \leq \epsilon\gamma/2.$$

Here $\|v_t - v^*\| \leq 2$, $\tau = \epsilon\gamma/8 < 1$, and $\sqrt{T} \geq 10/(\epsilon\gamma)$. Jensen's inequality gives the same hinge bound for $\bar{v} = T^{-1} \sum_t v_t$. Every classification mistake of $\text{sign}\langle \bar{v}, z \rangle$ incurs hinge loss at least γ , including a zero score when the true label is negative. Thus its error is at most $\epsilon/2$. There are nT scalar queries. This proves a deterministic, adversarial-oracle guarantee stronger than the stated expected-error requirement. $\square$

F.10 Proof of Theorem 7.10

Proof of Theorem 7.10. Take logarithms of Theorem 7.6, $\text{dc}(H_A) \leq (m+1)(\lceil 2/\tau \rceil + 1)^u$. The high-dimension event gives $\ln \text{dc}(H_A) \geq \ln N - \ln \log M + O(1)$. Under the stated query budget the numerator is at least $\frac{1}{2} \ln N$ for large N . The upper bound is Theorem 5.1. All constants depend only on the displayed fixed parameters. $\square$

F.11 Proof of Corollary 7.11

Proof of Corollary 7.11. The published theorem gives a uniform-product rectangle of the displayed product fraction. For rational row and column distributions, duplicate each row and column according to integer denominator multiplicities. Approximate sign-rank does not increase: duplicate columns use the same embedding, and every distribution on copies pushes forward to a distribution on original columns; duplicate rows require no new representation. Apply the rectangle theorem to this duplicated total matrix. Projecting a monochromatic rectangle to the original indices retains its sign and can only increase its product-weighted mass. Finally approximate arbitrary weights by rational weights. The maximum over the finitely many rectangle product masses is continuous, so the lower bound persists. Apply Theorem 3.1 with $\rho = \rho_\eta$. $\square$

G Proofs for differentiable-learning comparisons

G.1 Proof of Proposition 8.2

Proof of Proposition 8.2. The first hidden layer computes $\sigma(x_i), \sigma(-x_i)$ for every coordinate. These give x_i by subtraction and the constant $c = \sigma(x_1) + \sigma(-x_1) = 1$ on the cube. For every negative point p , a second-layer unit computes

$$\sigma \left(\sum_i p_i x_i - (n-1)c \right).$$

At $x = p$ its value is one; at every other cube point the sum is at most $n-2$ and the unit is zero. Pass c through one additional unit and output c minus twice the sum of the negative-point units. Pad unused units by zero weights. There are at most N negative points, and counting all fully connected

weights gives the claimed S . The moment-curve proof in Theorem 4.1 still applies to the padded domain. Padding preserves every dimension lower bound by restriction; it preserves a rectangle lower bound up to a factor two, because either padding carries at least half the point mass or the original domain does. $\square$

G.2 Proof of Proposition 8.3

Proof of Proposition 8.3. Given an oracle function, generate the learner's independent uniform point samples on ℓ_0 and label them by its oracle values at the corresponding incidence encodings. Run the learner without giving it a key. Draw an independent uniform test point on ℓ_0 , query its label, and output one if the predictor agrees. In the PRF case acceptance is at least $1/2 + \gamma(n)$. In the random-function case, conditional on the training transcript, a previously unqueried point has an independent fair label. A fresh test point repeats a training point with probability at most $q(n)/N$. Acceptance is therefore at most $1/2 + q(n)/(2N)$. The simulation and one output evaluation are polynomial-time. Since $N = 2^{(n-1)/3}$, the collision term is negligible, contradicting PRF security for inverse-polynomial γ . $\square$

H Proofs of the resource barriers

H.1 Proof of Theorem 10.1

Proof of Theorem 10.1. Let $K = 2N^3$, $b_N = \lceil \log_2(4K) \rceil$, and $d_N = \lfloor N/(128b_N) \rfloor$. For sufficiently large N , the counting bound in Theorem 4.1 guarantees a flip matrix of sign-rank greater than d_N : indeed $\log_2(4eK) \leq 2b_N$, so

$$2Kd_N \log_2(4eK) \leq N^4/16 < I.$$

Select the lexicographically first such matrix. This selection is computable. For a proposed matrix A , the predicate $\text{signrank}(A) \leq d_N$ is the truth of the finite real sentence

$$\exists U \in \mathbb{R}^{K \times d_N}, V \in \mathbb{R}^{d_N \times K} \quad \bigwedge_{i,j} A_{ij} \sum_{s=1}^{d_N} U_{is} V_{sj} > 0.$$

Decision of finite polynomial real sentences is the classical real-closed-field decision theorem. A constructive treatment of this decidability result is given by Mahboubi and Cohen [29]. Enumerate the finitely many flip tables and decide the displayed sentence for each until the first failure. The counting argument ensures termination. Small N can be handled by any fixed table.

A single fixed program performs this generation and then runs the finite rational rectangle-mixture algorithm. Encode N in $O(\log N)$ bits. For each fixed ϵ , run at a fixed rational accuracy below ϵ , whose description length is constant with respect to N . Conservative rational ceilings can replace the logarithmic constants in Theorem 3.1, making every step an ordinary finite computation. The table generation happens before the first SQ; it has no query cost. The learner therefore has $m = O(\log K)$ and constant positive tolerance, while the chosen class has $\text{dc} > d_N = \Omega(N/\log N)$. Its program description has size $O(\log N)$. The generated table occupies far more space, which this interpretation does not charge. This contradicts the displayed polynomial implication. $\square$

H.2 Proof of Proposition 10.2

Proof of Proposition 10.2. Fix a point prior μ . Apply the finite separation argument of Lemma 2.1 to the transposed matrix, obtaining a distribution λ over monochromatic rectangles $S_R \times T_R$ with

$$\forall h, \quad \mathbb{E}_R[\mathbf{1}_{T_R}(h)\mu(S_R)] \geq r, \quad r = \text{rect}(A).$$

In the Euclidean space indexed by rectangles, put

$$a_h(R) = c_R \sqrt{\lambda_R} \mathbf{1}_{T_R}(h), \quad b_x(R) = \sqrt{\lambda_R} \mathbf{1}_{S_R}(x).$$

Their norms are at most one, and their inner product is A_{hx} times the probability that R covers (h, x) . These products may be zero, so this alone is not a strict representation. Add an orthogonal strict unit-vector representation scaled by $\sqrt{\delta}$, and scale the displayed vectors by $\sqrt{1 - \delta}$. Every resulting product is strict with the required sign, and its average absolute value in each row is at least $(1 - \delta)r$. Pad row norms and point norms separately in orthogonal private coordinates to make them one without changing any cross inner products. Letting $\delta \downarrow 0$ proves $m_{\text{avg}}(A) \geq r$.

For the VC bound, suppose v points are shattered and use their uniform point prior. In any strict unit-vector representation, select one row h_σ for each sign pattern $\sigma \in \{-1, +1\}^v$. If the minimum row-average margin is a , then

$$va \leq \sum_{i=1}^v \sigma_i \langle v_{h_\sigma}, u_i \rangle \leq \left\| \sum_{i=1}^v \sigma_i u_i \right\|.$$

Averaging the square of the last norm over independent fair signs gives v , because cross terms cancel. Some sign pattern has norm at most $\sqrt{v}$, so $a \leq 1/\sqrt{v}$. Take the supremum over representations and the infimum over priors. For $v = 0$ the inequality is immediate. This proves the chain, including the strict-sign correction. $\square$

H.3 Proof of Proposition 10.3

Proof of Proposition 10.3. Condition on the sampled point indices, their observed labels, and the learner's random coins. The label at any unobserved point of ℓ_0 remains an independent fair bit. The learner's prediction there has conditional error $1/2$. A fixed point is unobserved with probability $(1 - 1/N)^T$. Average over the N points to get the first inequality. The second follows by induction from $(1 - z)^T \geq 1 - Tz$ for $0 \leq z \leq 1$. Rearranging proves the sample requirement. For the final claim, $T > N/2$ and $S \geq 2$ give $TS \geq N$, whereas $\text{dc}(H_A) \leq 2N + 1 \leq 3N$ for every table. $\square$

I Comparison with randomized statistical dimension

The rectangle theorem also bounds Feldman's randomized statistical dimension. The main learner does not use this characterization. For the search task $\mathcal{Z}_\eta$, an instance is a realizable labeled distribution $P_{D,h}$; an allowed solution is a predictor g with loss at most η . Fix any reference distribution P_0 on $Z = X \times \{-1, +1\}$, not necessarily realizable or fair-labeled. For a law $\mathcal{P}$ on predictors, put

$$\mathcal{B}(\mathcal{P}) = \{j : \Pr_{g \sim \mathcal{P}} [g \text{ is } \eta\text{-good for } j] < 1/4\}.$$

A query distribution covers this bad set at level $1/d$ when every j in it is distinguished from P_0 by more than κ with probability at least $1/d$. Taking the best $\mathcal{P}$ and query law, and the worst P_0 , is the covering formulation of Feldman's $\text{RSD}_{\kappa_1}(\mathcal{Z}_\eta, \kappa, 1/4)$.

Theorem I.1 (Rectangle bound for randomized statistical dimension). *Suppose H is finite and nonempty, $\rho \geq 1$, and $\text{rect}(H) \geq 1/\rho$. Let $L = \lceil \rho \ln(4/\eta) \rceil$ and $\kappa = \eta/(4L)$, where $0 < \eta < 1/4$. Then*

$$\text{RSD}_{\kappa_1}(\mathcal{Z}_\eta, \kappa, 1/4) \leq 4L.$$

This statement allows arbitrary target-dependent marginals and any reference P_0 .

Proof. Fix a finitely supported prior π on instances. It induces a prior ν on hypotheses. Let $\mathcal{Q}$ be the finite query family $q_{S,c}(x, y) = \mathbf{1}_S(x)\mathbf{1}_{y \neq c}$. Set

$$\beta = \max_{q \in \mathcal{Q}} \Pr_{j \sim \pi} [|P_j q - P_0 q| > \kappa].$$

Peel L rectangles using the prior-averaged example marginal restricted to the current unpeeled set U , and the unchanged hypothesis prior ν . Empty residual mass permits immediate completion. The rectangle condition gives both

$$\nu(T) \geq r, \quad \mathbb{E}_j D_j(S) \geq r \mathbb{E}_j D_j(U).$$

Suppose first $\beta < r$. On every instance with target in T , the mismatch query has expectation zero. Therefore $P_0 q \leq \kappa$: otherwise a prior mass at least r would be distinguished. Apart from a prior mass at most β , we then have $P_j q \leq 2\kappa$.

After L peels, the expected residual mass is at most $(1 - r)^L \leq \eta/4$. Markov's inequality shows that at least half the prior puts residual mass at most $\eta/2$. Outside a union of exceptional sets of total mass at most $L\beta$, each of the L wrong-color masses is at most 2κ . The predictor assembled from the rectangle colors and an arbitrary residual label consequently has loss at most $2L\kappa + \eta/2 = \eta$ for prior mass at least $1/2 - L\beta$. If $\beta \geq r$, then $L\beta \geq L/\rho \geq \ln(4/\eta) > 1/2$, so the following inequality remains true trivially:

$$\max_g \Pr_{\pi}[g \text{ good}] + L \max_{q \in \mathcal{Q}} \Pr_{\pi}[q \text{ distinguishes}] \geq 1/2. \quad (\text{I.1})$$

Apply Lemma 2.1 to the action pairs (g, q) , whose payoff on instance j is

$$\mathbf{1}_{g \text{ good for } j} + L \mathbf{1}_{q \text{ distinguishes } j}.$$

It gives laws $\mathcal{P}, \mathcal{Q}'$ satisfying

$$\Pr_{g \sim \mathcal{P}}[g \text{ good for } j] + L \Pr_{q \sim \mathcal{Q}'}[q \text{ distinguishes } j] \geq 1/2 \quad \text{for every } j.$$

On $\mathcal{B}(\mathcal{P})$, the second probability is greater than $1/(4L)$. This proves the required coverage.

The continuum of possible marginals creates no compactness gap in this minimax step. Both the predictor set and $\mathcal{Q}$ are finite. An instance is relevant only through the finite vector of Boolean good-solution and distinguishing-query indicators. There are finitely many such types. Choose one representative of each realized type and apply the finite argument to them. Its inequalities then hold for all original instances. No discretization in total variation and no factor polynomial in $|X|$ is introduced. $\square$

Comparison with Feldman's characterization. In Feldman [15, Theorem 4.8], the randomized upper bound, in the notation used here, is

$$\text{RQC}(\mathcal{Z}, \text{STAT}(\kappa/3), \alpha - \delta) = O\left(d \frac{R_{\text{KL}}(\mathcal{D})}{\kappa^2} \log \frac{R_{\text{KL}}(\mathcal{D})}{\kappa\delta}\right), \quad d = \text{RSD}_{\kappa_1}(\mathcal{Z}, \kappa, \alpha). \quad (\text{I.2})$$

The displayed asymptotic expression uses the usual positive-radius conventions; a ceiling and a maximum with one handle degenerate values. Here $R_{\text{KL}}(\mathcal{D}) = \inf_Q \sup_{P \in \mathcal{D}} \text{KL}(P \| Q)$. Taking uniform Q on Z proves $R_{\text{KL}} \leq \ln |Z|$: expand $\text{KL}(P \| Q) = \ln |Z| + \sum_z P(z) \ln P(z)$ and use $\ln P(z) \leq 0$. Thus the relevant domain term is $\ln(2|X|)$, not $|X|$ or the logarithm of a discretized instance count. The deterministic theorem has a different set-cover logarithm; it must not be substituted here.

For clarity, the mechanism behind (I.2) does not depend on a hidden finite-instance enumeration. Maintain a reference Q starting uniform on Z . A query separated from the true P by κ supplies a

bounded direction with a signed expectation gap of order κ . For $f \in [-1, 1]$ and update $Q'(z) \propto Q(z)e^{\lambda f(z)}$, direct expansion gives

$$\text{KL}(P\|Q') - \text{KL}(P\|Q) = -\lambda \mathbb{E}_P f + \log \mathbb{E}_Q e^{\lambda f} \leq -\lambda(\mathbb{E}_P f - \mathbb{E}_Q f) + \lambda^2/2.$$

The final inequality follows by twice differentiating $\log \mathbb{E}_Q e^{\lambda f}$: its second derivative is a variance of a variable in $[-1, 1]$, at most one. Choosing λ proportional to κ decreases KL by $\Omega(\kappa^2)$. There are therefore $O(\ln |Z|/\kappa^2)$ updates. At each reference, sampling $O(d \ln(T/\delta))$ covering queries prevents an undetected bad instance over T updates, by $(1 - 1/d)^s \leq e^{-s/d}$ and a union bound. This proves the parameter dependence and extends the argument to all realizable marginals here.

At fairness $1/4$ and constant $\delta < 1/4$, this yields constant success probability. Fresh repetitions and a single loss query to verify each candidate amplify success with $O(\ln(1/\delta'))$ repetitions and loss slack $O(\kappa)$. This reasoning remains valid for history-dependent oracles by conditioning on earlier runs. The verifiable-search Theorem 4.11 in Feldman [15] is stated using a different statistical dimension over references admitting no good solution; its dimension must not be identified automatically with the fairness- $1/4$ quantity above. In either route verification does not erase the R_{KL}/κ^2 term of that generic simulation. The direct learner of Theorem 3.1 gives the sharper bound without it.

J A finite-bit convex program for the cap implementation

This appendix proves the weakly polynomial implementation needed in Theorem 5.1. It does not require a claim that an exact radial-isotropic matrix has rational entries. Approximate Forster computation is established prior work; see Diakonikolas et al. [11] for stronger algorithms. We verify the particular general-position convex program and the bit bounds needed here.

Let $k > d$ and let $v_1, \dots, v_k \in \mathbb{Q}^d$ be in linear general position. Assume the numerator and denominator of every coordinate have absolute value at most 2^b , with nonzero denominators. Let Δ_I be the determinant of the rows indexed by a d -subset I . Every Δ_I is nonzero. Cauchy–Binet gives the convex function

$$f(t) = \log \det \left(\sum_i e^{t_i} v_i v_i^\top \right) - \frac{d}{k} \sum_i t_i = \log \sum_{|I|=d} \Delta_I^2 e^{\sum_{i \in I} t_i} - \frac{d}{k} \sum_i t_i. \quad (\text{J.1})$$

The identity follows by applying the determinant expansion to the product of the $d \times k$ column matrix with entries $e^{t_i/2} v_i$ and its transpose; only distinct d -subsets remain. Differentiating the log-sum-exp expression shows that its Hessian is the covariance matrix of the incidence vector $\mathbf{1}_I$ under the displayed positive weights. It is positive semidefinite and has operator norm at most $\mathbb{E} \|\mathbf{1}_I\|^2 = d$. Also $\|\nabla f\| \leq 2\sqrt{d}$. Thus f is convex, d -smooth, and invariant under adding a constant to all coordinates.

We give a polynomial radius bound on the zero-sum gauge. Write $a_- = \min_I \log \Delta_I^2$ and $a_+ = \max_I \log \Delta_I^2$. For $t_1 \geq \dots \geq t_k$ with $\sum t_i = 0$, put $s = t_1 + \dots + t_d$. The average of the largest $d-1$ of $t_2, \dots, t_k$ is at least their overall average; hence

$$s \geq t_1 \frac{k-d}{k-1} \geq \frac{t_1}{k-1}.$$

Likewise the largest d of $t_1, \dots, t_{k-1}$ have average at least $-t_k/(k-1)$, giving $s \geq -dt_k/(k-1)$. Therefore $s \geq \|t\|_\infty/(k-1)$, and

$$f(t) \geq a_- + \frac{\|t\|_\infty}{k-1}, \quad f(0) \leq a_+ + \log \binom{k}{d}.$$

Sublevel sets on the gauge are compact. A minimizer exists and has infinity norm at most $(k-1)(a_+ - a_- + \log \binom{k}{d})$.

The determinant bounds are explicit. A common denominator for a $d \times d$ determinant is the product of its d^2 coordinate denominators, so

$$2^{-bd^2} \leq |\Delta_I| \leq d!2^{bd}.$$

Consequently the integer

$$R = 1 + (k-1)(2bd^2 + 2bd + 2d\lceil \log_2(d+1) \rceil + d\lceil \log_2(k+1) \rceil)$$

exceeds the minimizer radius. Work on the rational polytope $Q = \{t : \sum t_i = 0, |t_i| \leq 2R\}$.

For completeness, a deliberately conservative projected-subgradient implementation suffices. Set

$$\delta = \frac{d}{512k^2}, \quad a = \frac{\delta}{36d}, \quad \xi = \frac{\delta}{32Rk}, \quad T = \left\lceil \frac{4R^2k}{a\delta} \right\rceil.$$

At each step compute a rational approximation $\tilde{g}$ of ∇f with Euclidean error at most ξ , and project $t - a\tilde{g}$ onto Q . The projection is rational and polynomial-time: it is $\text{clip}(z_i - \lambda, -2R, 2R)$ in each coordinate, with λ chosen to make the sum zero. The sum is a continuous piecewise-linear decreasing function; sorting its $2k$ breakpoints and solving on the containing interval finds λ . The projection inequality follows directly from its normal-cone signs: for every $u \in Q$, $\langle z - \Pi_Q z, u - \Pi_Q z \rangle \leq 0$, so $\|\Pi_Q z - u\| \leq \|z - u\|$.

Let t^* be a minimizer with $\|t^*\| \leq R\sqrt{k}$, and start at zero. The standard distance expansion, supplied here to account for error, gives

$$\frac{1}{T} \sum_{s < T} (f(t_s) - f(t^*)) \leq \frac{R^2k}{2aT} + \frac{a}{2}(2\sqrt{d} + \xi)^2 + 4R\sqrt{k}\xi.$$

Indeed convexity bounds $f(t_s) - f(t^*)$ by $\langle \nabla f(t_s), t_s - t^* \rangle$, the gradient error contributes at most $4R\sqrt{k}\xi$, and squared distances telescope. Since $\xi \leq \sqrt{d}$, each of the three terms is at most $\delta/8$. Convexity gives $f(\bar{t}) - \min f < \delta$ for the rational average $\bar{t}$.

Smoothness on the full space implies

$$\|\nabla f(\bar{t})\|^2 \leq 2d(f(\bar{t}) - \min f) < \frac{d^2}{256k^2} :$$

apply the quadratic upper bound to $\bar{t} - \nabla f(\bar{t})/d$ and compare with the global minimum. Put $S = \sum_i e^{\bar{t}_i} v_i v_i^\top$. Differentiating the first expression in (J.1) yields

$$\frac{15d}{16k} < \ell_i := e^{\bar{t}_i} v_i^\top S^{-1} v_i < \frac{17d}{16k}.$$

For any factor T_0 with $T_0 S T_0^\top = I$, the normalized row covariance satisfies

$$\frac{16}{17d} I \preceq \frac{1}{k} \sum_i \frac{T_0 v_i v_i^\top T_0^\top}{\|T_0 v_i\|^2} \preceq \frac{16}{15d} I.$$

To see this, write each summand as $e^{\bar{t}_i} T_0 v_i v_i^\top T_0^\top / \ell_i$ and use the preceding bounds together with their unnormalized sum I .

All calculations can be performed to the required accuracy with polynomially many bits, rather than an unspecified exact-real oracle. For fixed d , the number of terms $\binom{k}{d}$ in (J.1) is polynomial. The gradient is a ratio of sums of the nonnegative numbers $\Delta_I^2 \exp(\sum_{i \in I} t_i)$. Their exponent arguments have absolute value at most $2dR$, and their rational coefficients have polynomial bit length by the determinant bounds. Exponentials on this bounded range admit rational interval approximations

with bit complexity polynomial in $R, b, k, \log(1/\xi)$: for nonnegative arguments use the Taylor series and its positive tail bound, and for negative arguments take reciprocals. The elementary inequality $j! \geq (j/e)^j$, obtained by integrating $\log x$, bounds the number of Taylor terms by a polynomial in the argument bound and target precision. Positive sums have no cancellation; the denominator has the explicit lower bound $2^{-2bd^2} e^{-2dR}$. Compute each gradient coordinate to error at most $\xi/(2k)$ and round it to one fixed dyadic grid of spacing at most $\xi/(2k)$. The total coordinate error is at most ξ/k , giving the required Euclidean error. These grid values have polynomial bit length independent of the iteration index. Exact rational projection introduces only the denominators of the fixed step size and a divisor at most k at each iteration, so projection and averaging retain polynomial bit length over the polynomial number of iterations. Thus the preceding inexact-gradient implementation has a polynomial bit-operation bound for fixed d .

Finally, approximate T_0 by an invertible rational matrix T . This also requires only polynomial precision. The bounds

$$\lambda_{\max}(V^\top V) \leq kd 2^{2b}, \quad \lambda_{\min}(V^\top V) \geq \frac{2^{-2bd^2}}{(kd 2^{2b})^{d-1}}$$

follow from trace, determinant, and Cauchy–Binet. Since $|\bar{t}_i| \leq 2R$, multiplying these bounds by $e^{\pm 2R}$ bounds the eigenvalues of S . Their logarithms have polynomial size. Interval Cholesky factorization followed by inversion, approximating each square root by rational bisection, consequently computes T_0 to $2^{-q(k,b,R)}$ operator error in polynomial time for a sufficiently large fixed polynomial q . This last assertion can also be checked directly: Cholesky pivots are bounded below by the smallest eigenvalue, all divisions have polynomial logarithmic conditioning, and there are only $O(d^3)$ operations. If $\|a - b\| \leq \|a\|/2$, then

$$\|a/\|a\| - b/\|b\|\| \leq 2\|a - b\|/\|a\|.$$

The displayed eigenvalue bounds and the nonzero rational input coordinates therefore give an explicit polynomial precision at which the normalized covariance changes by less than $1/(8d)$. Its lower eigenvalue remains above $3/(4d)$. For rational T , this final statement is independently decidable by rational arithmetic: the covariance is the sum of $Tv_i v_i^\top T^\top / (v_i^\top T^\top T v_i)$, and positivity above $3I/(4d)$ is checked using leading principal minors. This completes the finite-bit construction.

SQ replies themselves need no unlimited precision in the computational corollary. Request half the advertised tolerance and round the reply to a rational grid of spacing at most that tolerance. The effective error remains within the advertised tolerance. This changes only an absolute constant and uses $O(\log(1/\tau))$ reply bits. Each cap test is a fixed-degree rational inequality; the unpeeled set is represented by the negations of earlier peeling tests. No table-dependent advice beyond the stated input is hidden in this computation.

K The cube concentration calculation used for non-necessity

Let f on the uniform n -cube change by at most one when one coordinate is flipped. Reveal the coordinates successively and write the Doob differences as D_i . Conditional on the first $i-1$ coordinates, the two conditional expectations after revealing coordinate i differ by at most one: couple the remaining coordinates identically and use the bounded-difference hypothesis. Thus D_i is conditionally centered and lies in an interval of length at most one.

For any centered random variable Z in an interval of length one, let $F(\lambda) = \log \mathbb{E} e^{\lambda Z}$. Differentiation of the finite sum gives $F''(\lambda)$ equal to the variance under the exponentially tilted law. A variable $Y \in [a, b]$ has variance at most $(b-a)^2/4$: from $(Y-a)(b-Y) \geq 0$ obtain $\text{Var}(Y) \leq (\mathbb{E}Y - a)(b - \mathbb{E}Y) \leq (b-a)^2/4$. Hence $F'' \leq 1/4$, and integrating twice using $F(0) = F'(0) = 0$ yields $F(\lambda) \leq \lambda^2/8$ for either sign of λ .

Iterating conditional expectation gives

$$\mathbb{E} \exp\{\lambda(f - \mathbb{E}f)\} \leq \exp(n\lambda^2/8).$$

Markov's inequality and $\lambda = 4a/n$ imply $\Pr[f - \mathbb{E}f \geq a] \leq e^{-2a^2/n}$. Apply the same calculation to $-f$ for the lower tail. All distributions here are finite, so differentiation and conditional expectations require no limiting justification.

L Additional geometric bounds

Proposition L.1 (Independent constant rectangle bound). *Let c_d be the normalized surface measure of the cap of Section 5.1 on S^{d-1} , $d \geq 2$. Every matrix F of sign-rank at most d satisfies $\text{rect}(F) \geq c_d/(4d)$. If F has no negative 2×2 , every monotone flip A satisfies $\text{rect}(A) \geq c_d/(8d)$. In particular, for $d = 6$ this is greater than 2^{-18} .*

Proof. First take a uniform prior on $k > d$ row copies. Perturb their representing vectors, if necessary, so every d are independent while preserving the finitely many strict signs. The full minimization argument in Theorem 7.4 produces an invertible transform whose normalized row vectors are isotropic. Apply the inverse transpose to the point vectors and normalize; all signs persist.

For each uniform random direction u , the two sets $T_+(u) = \{v : \langle u, v \rangle > \alpha\}$ and $T_-(u) = \{v : \langle u, v \rangle < -\alpha\}$ have combined prior mass at least $1/(2d)$. The rectangles $C_u \times T_+(u)$ and $C_u \times T_-(u)$ have constant signs. For any fixed point marginal μ , rotational invariance gives $\mathbb{E}_u \mu(C_u) = c_d$, because each unit point lies in a random cap with probability c_d . Therefore the expected sum of their product masses is at least $c_d/(2d)$, and some single rectangle has mass at least $c_d/(4d)$.

Apply Lemma 2.2, repeating all row copies further so their number exceeds d . Representations of identical copies may differ, but their signs agree with the original entries; projection therefore preserves monochromaticity. A negative rectangle has a singleton side under the additional assumption on F . Split its other side according to the new signs in A ; one color retains at least half its product mass. Positive rectangles survive unchanged. This proves the second assertion.

For the numerical estimate at $d = 6$, project the cap onto its last five coordinates. Its image is the five-dimensional ball of radius $\alpha = 1/\sqrt{12}$, and the surface-area Jacobian is at least one. The volume of the unit five-ball is $8\pi^2/15$ and the area of S^5 is π^3 , obtained by polar integration of the Gaussian integral. Thus

$$c_6 \geq \frac{8}{15\pi 12^{5/2}} > \frac{8}{15 \cdot 4 \cdot 576} = \frac{1}{4320}, \quad \frac{c_6}{48} > \frac{1}{207360} > 2^{-18}.$$

This independent constant, rather than the sharper imported one, is already sufficient for Theorems 3.1–4.2 to refute every polynomial dimension implication. $\square$

Proposition L.2 (Uniform-product rectangles in a projective plane). *For the point–line incidence sign matrix of $\text{PG}(2, q)$, let $K = q^2 + q + 1$ and $k = q + 1$. Under uniform distributions on both sides, the largest monochromatic-rectangle mass lies between*

$$\frac{1}{8(q+1)} \quad \text{and} \quad \frac{q}{(q+1)^2}.$$

It is therefore $\Theta(1/q)$.

Proof. Let P be the incidence matrix. As proved in Theorem 7.5, $P\mathbf{1} = k\mathbf{1}$ and $P^T P = qI + \mathbf{1}\mathbf{1}^T$. Thus $C = P - (k/K)\mathbf{1}\mathbf{1}^T$ has operator norm $\sqrt{q}$. A nonincident rectangle with side sizes a, b satisfies

$$\frac{k}{K}ab = |\mathbf{1}_S^T C \mathbf{1}_T| \leq \sqrt{q}\sqrt{ab},$$

so its mass is at most q/k^2 . An all-incident rectangle has a singleton side and size at most k . Its mass k/K^2 is also at most q/k^2 : use $K \geq qk$ and $q^3 \geq k$ for $q \geq 2$ to obtain $qK^2 \geq q^3k^2 \geq k^3$.

For the lower bound, choose $s = \lfloor K/(2k) \rfloor$ lines. Their union has at most $sk \leq K/2$ points, leaving at least $K/2$ points incident with none of them. The resulting rectangle has mass at least $s/(2K)$. Since $K/(2k) \geq 1$, $s \geq K/(4k)$, proving the lower bound. $\square$

M Experiments and reproducibility

All experiments use CPU computation. The public key is

8959fe4a6b404807275400d0a95800e62d0b
119c2f8e0061d0ff5d241b229c78

with the two lines concatenated. The experimental flip bit is the low bit of the first byte of HMAC-SHA256 on the key and the message consisting of `SQDC4`, a zero byte, and the unsigned 64-bit big-endian tuple (N, a, b, x, y) . Off-incidence entries remain positive. The random-search seed is 202609104. This specifies a reproducible keyed-hash instantiation, not a proof of the subexponential security assumed in Theorem 6.2.

Oracle and distribution access. The learners receive only the entry evaluator and SQ answers; the simulator alone knows the chosen target and rational distribution. Every query records its support mask, exact expectation, returned rational answer and public control state. The adversarial policy chooses a tolerance endpoint using the complete public transcript and the current query. For threshold decisions it chooses an endpoint that favors premature stopping, ambiguous median comparisons, a high-error candidate acceptance, or continued search when available, according to the explicit priority rules in the scripts. The cap marginal policy cannot inspect a private row side or color not revealed by the query. The policy is a concrete greedy adversary, not an optimizer over future private coins. Exact-oracle runs are reported separately in the raw outputs.

Structured distributions include uniform template points, actual negative points, one heavy negative point with a small tail, and mixed positive/negative supports. The searches use random restarts followed by coordinate changes of rational weights, with support at most 20. The order-median sweep uses three restarts and four mutations; the cap sweep uses two restarts and three mutations. Reporting uses fresh learner seeds. Every recorded error is recomputed against the actual target, and every recorded proper output is checked for class membership.

Succinct order-median learner. The script `seeded_fast.py` executes the proper algorithm of Theorem 5.4 for $N = 2, \dots, 8, 16, 32$. It does not construct a truth table or enumerate all domain points inside the learner. The simulator represents D by its finite support. The stated accuracy is $\epsilon = 1/10$, and the exact-rational experiments use effective tolerance $\zeta = \epsilon/4096 = 1/40960$. The finite-bit theorem requests raw tolerance $\zeta/2$ to reserve the other half for response rounding; these runs have no additional rounding error.

N	Runs	Largest error	Most queries	Max. wall (ms)	Slope-search runs
2	16	0.0010000000	8	0.354	0
3	36	0.0010000000	11	0.445	0
4	36	0.0143149285	12	0.253	0
5	36	0.0243217961	14	0.254	0
6	36	0.0010000000	28	0.377	2
7	36	0.0010000000	15	0.239	0
8	36	0.0010000000	16	3.155	0
16	36	0.0010000000	52	0.955	12
32	36	0.0010000000	89	1.394	16

Error and wall-time entries are rounded upward. Wall time includes the exact finite-support oracle, not the cost of a physical statistical oracle. Raw output also records simulator time separately. These are empirical maxima and single-run timings, not asymptotic runtime estimates. All 304 runs passed 4,798 exact query-answer checks; all 304 outputs were proper. The 30 slope-search runs specifically exercise the order comparison used to resolve the all-light case.

The small grid sweep cannot reach the large-star sampling branch because its conservative threshold is at least $128/\epsilon = 1280$. A separate unit test therefore uses $N = 4096$ and the point $p = (1, N + 1)$, whose star has 4096 lines. Exactly 1968 of these keyed rows are negative at p , exceeding the required local quarter fraction. On 20 learner seeds, a distribution putting mass .7 at p and .3 at a positive outside point produced a proper zero-error answer. Full transcripts are saved. This tests the large-star branch at one verified star; it does not claim that the entire $N = 4096$ table was generated or tested for regularity.

Keyed Proper RECTIFY. The script `run_keyed_caps.py` runs the proper cap learner on the same keyed family for $N = 2, \dots, 8$, using only the public-history endpoint oracle. It uses outer accuracy $1/10$, rectangle-phase accuracy $1/40$, $\rho = 2^{20}$, and tolerance $1/23226329280$. A numerical radial-isotropic optimizer proposes a finite catalog of directions and rectangles. Before that catalog is used, the integer truth table verifies every rectangle and the exact pointwise coverage of the discrete mixture. Thus the implemented coverage guarantee does not rely on rounding a numerical covariance residual in the favorable direction.

N	Runs	Largest error	Most rounds	Most queries	Max. wall (s)
2	8	.0062500001	63	133	.010
3	8	.0062500001	75	159	.031
4	8	.0085570466	198	409	.105
5	8	.0062500001	393	798	.203
6	8	.0062500001	645	1308	.462
7	8	.0062500001	2881	5779	1.446
8	8	.0062500001	749	1521	3.838

The 56 recorded runs passed 28,262 exact query-answer checks. Their 499 catalogs passed 231,150 monochromatic-rectangle checks and 228,684 exact pointwise coverage checks. Every returned hypothesis belonged to the class. The entire keyed-cap experiment, including search, took 63.35 seconds in the recorded environment. Query types and the peel, elimination and rejection counts appear in the aggregate CSV and full transcripts. The much smaller query count of the order-median method is expected: it uses the grid order rather than sampling from a conservative generic rectangle law.

Subgrid counts and elementary checks. The following exact counts enumerate the independent incidence positions. The rank threshold from Lemma 6.1 is zero at all these small scales, so its low-rank event is empty. This table checks the construction and constants, not the asymptotic sign-rank lower bound experimentally.

k	$K_k = 2k^3$	I_k	D_k	$k^4/2$
2	16	23	0	8
3	54	126	0	40.5
4	128	412	0	128
5	250	1025	0	312.5
6	432	2151	0	648

The script `algebra_and_stress.py` passed 280 exact rational endpoint checks for the new median, split and orientation inequalities. It also exhausts all label patterns and all sample-index sequences for $N \in \{2, 3, 4\}$ and $T \in \{0, 1, 2, 3\}$: memorization with a constant fallback on unseen points achieves exactly $\frac{1}{2}(1 - 1/N)^T$ on average in each of the 12 cases. The affine certificate script additionally checks the incidence and Gram identities over five prime fields and the corresponding success-probability substitution. The experiments concern statistical-query algorithms.

Scope of verification. The proofs establish the universal statements, subject to explicitly cited imported theorems and the stated cryptographic assumption. The recorded response, membership and coverage checks are exact finite certificates; numerical solver convergence and adversarial-search global optimality are not certified. Keyed-hash experiments test an implementation, not cryptographic indistinguishability or uniform empirical performance over all keys. Build commands, software versions, seeds, sources and raw rational outputs accompany the manuscript.